

North American Bat Monitoring Program in SE Alaska

Data Summary and Activity Trend Analyses: 2016 - 2025

Camila Hurtado

Taylor Hart

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1 Executive Summary

The North American Bat Monitoring (NABat) program is a multi-agency initiative administered by the US Geological Survey (USGS). In 2024, the North by Northwest (NNW) Bat Hub was formed by combining the former Alberta Hub and the BC and Southeast Alaska (BC-SE_AK) Hub. The NNW Bat Hub leads the NABat program within S.E. Alaska, with key support from state representatives. In 2025, 6 grid cells in S.E. Alaska were monitored. Results indicate that bat populations across S.E. Alaska remain stable, with some key species showing small positive trends, including Little Brown Myotis and Eastern Red Bat. Given growing threats to bat populations in BC, continued monitoring is essential to detect any significant changes.

2 Introduction

2.1 Overview of NABat and the NNW Bat Hub

North American Bat Monitoring Program (NABat) is a large-scale coordinated effort to monitor bat species to address gaps in knowledge and lack of long-term studies across North America (Loeb et al. 2015). The program is administered by the US Geological Survey (USGS) and implemented by the North by Northwest Bat (NNW) Hub in S.E Alaska, Alberta, and British Columbia.

NNW Bat Hub was established in 2024 with collaboration from the governments of Alaska, Alberta, and British Columbia. The hub was formed by combining two previous regional hubs: the Alberta Hub and the BC and Southeast Alaska Hub. NABat monitoring in Southeast Alaska is implemented by the Alaska Department of Fish and Game and US Forest Service.

2.2 Threats to bat populations in Alaska

Bat populations in Alaska face a wide range of threats including forestry practices, anthropogenic expansion, climate change, and disease. One of the most concerning threats is White-Nose Syndrome (WNS), a deadly fungal disease caused by the fungus *Pseudogymnoascus destructans* (*Pd*) that affects hibernating bats. As of 2026, WNS has been confirmed in 42 U.S. states, nine Canadian provinces, and the Northwest Territories (Figure 1: Segers (n.d.) ; U.S. Fish and Wildlife Service (USFWS) (n.d.-a)). The fungus, but not the disease, has been detected in 5 additional states and within British Columbia. Continued monitoring of bat populations in Alaska is critical as *Pd* and WNS continue to spread throughout the region.

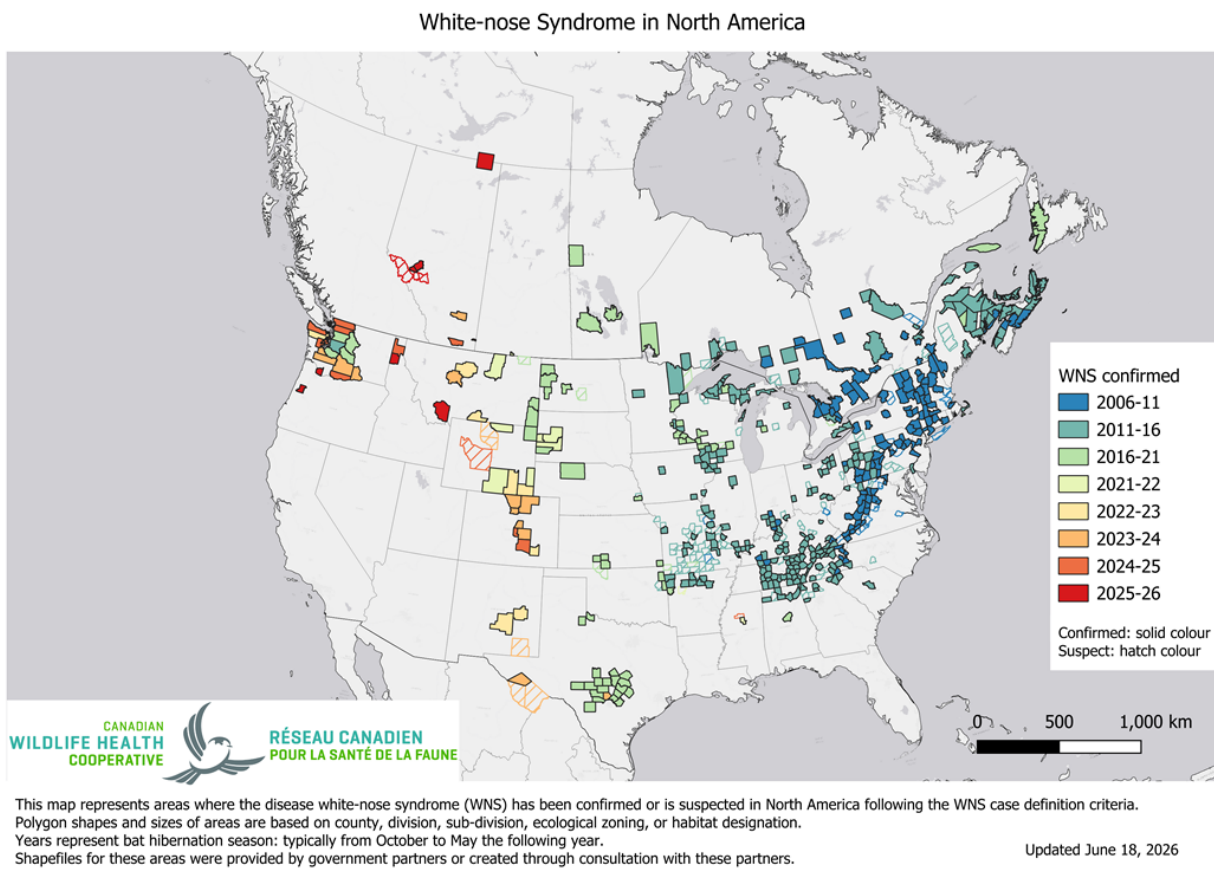


Figure 1: White-nose syndrome (WNS) and *Pseudogymnoascus destructans* (Pd) occurrence in The US and Canada. Source: CWHC-RCSF (2026)

2.3 Acoustic Monitoring of Bats

Acoustic recording is a cost-effective method for monitoring bat populations across large geographic and temporal scales. However, several limitations affect the conclusions that can be drawn from these data:

1. **Imperfect species identification** — Each bat species can produce a broad range of echolocation calls, corresponding to different flight behaviours and environmental features. Species with similar ecology and physiology often produce overlapping calls, particularly in areas with high clutter (obstacles), such as forested habitats. Although detectors are deployed to maximize recordings of diagnostic Search Phase calls, species identifiability will always vary depending on the local species assemblage.
2. **Variable detectability** — Bat species are not equally detectable acoustically. Larger, wide-ranging species produce louder calls that travel farther, whereas smaller species with quieter calls or specialized habitat use may be less reliably recorded. As a result, trend estimates cannot be produced for all species using acoustic methods alone.
3. **Data variability** — Many factors influence recording quality and bat activity, including weather, prey abundance and distribution, and local habitat. Additionally, stationary detectors may record an individual bat multiple times, meaning stationary acoustic data can only inform relative activity and occupancy.
4. **Processing limitations** — Auto-ID software enables efficient processing of large datasets, but is known to have high misclassification rates for bats, making manual verification necessary to ensure accurate results.

2.4 Statistical Models

Bat monitoring data can be analyzed using different statistical approaches, each with distinct assumptions and outputs. Activity trend models examine changes in relative acoustic activity over time, providing a straightforward measure of whether bat detections are increasing, decreasing, or remaining stable across the monitoring network.

Dynamic occupancy models take a different approach, estimating the probability that a species occupies a given site, while explicitly accounting for imperfect detection. Rather than treating a non-detection as an absence, these models distinguish between a species being truly absent versus simply going undetected, an important consideration given the variable detectability of bats discussed above. They also track colonization and extinction probabilities across sites over time, providing insight into range shifts and local population dynamics.

In 2025, both model types were run to examine differences in results and assess the complementary information each approach provides.

3 Methods

3.1 Data Collection

3.1.1 Stationary Surveys

Two to four stationary detectors were deployed across at least two quadrants per grid cell for a minimum of seven nights. Sites were selected to maximize species detection and recording quality across different habitat types (see AK report for details). Deployment occurred between mid-May and mid-July, after bats have emerged from hibernation but before young bats are volant. Sites and timing should be kept consistent between years to minimize variability in conditions affecting bat activity.

3.1.2 Transect Surveys

Where possible, driving transects were conducted following Loeb et al. (2015). Bats were recorded along the road by driving at ~32km/h with a microphone mounted vertically inside a magnetic pvc pipe holder on the vehicle's roof. Transects were 30-45 km long and were surveyed twice during the stationary deployment period whenever possible.

3.1.3 Landscape and Weather Covariates

Model covariates for the activity trend models and dynamic occupancy models were based on those established by Rae and Blejwas (2024) , with additional exploration of factors influencing colonization and extinction in the dynamic occupancy models. For the dynamic occupancy models we included distance to the nearest road, calculated as the distance from each survey location to the closest road feature. Road features for British Columbia were derived from the unpublished Human Footprint Index (HFI) layer currently being developed by the Geospatial Centre within Biodiversity Pathways, while those for Southeast Alaska were taken from the [Alaska Department of Transportation and Public Facilities \(AKDOT\) Roads](#) layer.

Temperature and relative humidity were recorded in each grid cell using onsite logging devices where possible; where loggers were unavailable, values were drawn from the nearest climate station. For sites in British Columbia, station records were obtained from the Pacific Climate Impacts Consortium (PCIC) Provincial Climate Data Set ([PCDS](#)), which aggregates and quality-controls observations from multiple provincial and federal networks. For sites in Southeast Alaska, station records were obtained using the meteostat library in python, which aggregates weather station records from multiple agencies. Additional weather covariates drawn from these station records included nightly mean wind speed (km/h) and cumulative precipitation since January 1 (mm). Degree days above 10°C since January 1 were calculated from daily temperature records, and nightly moon illumination (proportion illuminated × proportion of the night the moon was above the horizon) was calculated for each site-night.

Site-level covariates, including distance to clutter, percent clutter, distance to water, and water feature type, were recorded by grid leaders during deployment.

3.1.4 Manual Verification

In 2025 acoustic analysis mostly followed the standards set by WCSC in previous years (Rae and Blejwas 2024). This included running all files through two automatic classifiers, Kaleidoscope’s Bats of North America 5.4.0 classifier and Sonobat 3.0’s southeast Alaska classifier, to remove files that do not contain bat pulses and giving a preliminary bat species identification.

Manual vetting largely follows the protocol of Reichert et al. (2018) and previous analysis standards developed by WCSC. The key difference in the 2024 and 2025 acoustic analysis compared to previous years is that not all automatic species identifications were manually verified. Instead, based on findings from Stratton et al. (2022) , at least 50% of recordings were selected for manual verification.

Subsetting of for manual verification

Starting in 2024, the NNW Bat Hub implemented manual vetting of only a subset of all the bioacoustic data collected under the NABat program. Given the project’s large scale, manually vetting all recordings is not feasible long-term. To address this, the NNW Bat Hub, in collaboration with WCSC and Brian Paterson, developed procedures to ensure that selecting a subset of recordings for manual review minimizes biases in analytical results. These standards prioritize the accurate identification of rare and difficult-to-recognize species to maintain the integrity of models and analyses.

To ensure data accuracy, manual verification was conducted based on the number of recordings that received an autoID tag at each site. For sites with 200 or fewer tagged recordings, all non-noise recordings were manually reviewed. For sites with more than 200 tagged recordings, verification depended on the proportion of recordings assigned a species label. All tags for rare species (comprising 10% of prior years’ identifications) and commonly misidentified species were fully vetted. Specifically, all Hoary Bat (*Lasiurus cinereus*) and Silver-haired Bat (*Lasiyonicteris noctivagans*) were reviewed due to the frequent misclassification of noise files that are tagged as these species. Additionally, recordings of common species were randomly selected for verification until 50% of total recordings were reviewed. If fewer than 50% of recordings had species tags, all classifier-assigned tags were manually verified, and “NoID” recordings were randomly selected until 50% of all recordings were reviewed.

3.2 Statistical Analyses

3.2.1 Activity Trends

Statistical analyses followed closely what was established by WCSC and Dr. Carl Schwarz in 2021 (Rae and Blejwas 2024). This includes conducting species-specific trend analyses examining activity patterns in the data collected to date using the (negative binomial GLMM) models and the covariates identified in Table 1.

Table 1: Top-performing stationary models selected via AIC-based stepwise selection. Covariates are described as, Moon = moon illumination $\times$ proportion of night above horizon; Clutter = distance (m) to nearest object; Nearby Water = detector within 100m of water; RH = mean nightly relative humidity (%); Degree Days = cumulative degree days above 10°C since Jan 1. Source Rae and Blejwas (2024)

Species	Species Code	Top Performing Model Covariates
Hoary Bat	LACI	Distance to Clutter m ,Water Nearby,Nightly Max Temperature,Nightly Max RH
Silver-haired Bat	LANO	Percent Clutter,WaterDist,Nightly Mean Wind Speed
California Myotis	MYCA	Percent Clutter,WaterDist,Nightly Precipitation
Long-eared Myotis	MYEV	Distance to Clutter m ,Nightly Max Temperature,Nightly Min RH,jNight
Little Brown Myotis	MYLU	Percent Clutter,WaterDist,Nightly Max Temperature,Nightly Mean Wind Speed
Long-legged Myotis	MYVO	Nightly Precipitation,Nightly Mean Wind Speed
Yuma Myotis	MYYU	Percent Clutter,WaterDist,Nightly Max RH

Species were excluded from the analysis if model fit was not appropriate due to issues including under-dispersion or standard errors that were more than 3 times the estimate. Model results were also excluded if there was less than 150 recordings across all sample years. We conducted our statistical analyses on provincial and regional scales. For the regional analyses, we assigned all sampled grid cells into five geographic regions (Alaska, Southwestern, Southcentral, Southeastern, and Northern), each roughly corresponding to different species sets. Since transects are only conducted in roughly 50% of our grid cells on a typical year and each transect records fewer bats per night than stationary monitoring, we have elected not to conduct regional-scale transect trend analyses due to low sample sizes.

3.2.1.1 Use of Automated Identification Results in Stationary Trend Analyses

For stationary trend analyses, species identifications came from AutoID (Kaleidoscope Pro and Sonobat 3.0), not from manual verification. Manual review was used only to remove non-bat noise. Specifically, files that manual review flagged as noise were excluded, but where manual review suggested a different species than the AutoID label, the AutoID classification was retained.

This approach was first identified as necessary in 2023 (Rae and Lausen 2024), when analyses of manually verified data revealed a consistent positive trend across all species, this was interpreted as an artifact of reviewer learning over time rather than a true biological signal. As verifier experience increased across years, calls were increasingly assigned to specific species rather than left unidentified, producing an apparent increase in detections independent of actual bat activity. Because this observer effect is difficult to control for statistically, AutoID classifications are now used as the standard for all activity trend analyses, as automated classifiers apply consistent criteria across all

years. While AutoID is subject to some misclassification, the error rate is consistent across years and does not confound detection of genuine temporal trends.

Trend analyses are presented only for species confirmed present through manual verification and for which models fully converged.

3.2.2 Dynamic Occupancy

This analysis used nightly acoustic detection/non-detection data from passive bat detectors deployed at GRTS-selected sites in British Columbia (BC) as part of the NABat long-term monitoring program. Data were filtered to the 2017–2025 period, and analysis used an every-other-night subsampling protocol to reduce temporal autocorrelation between consecutive nights, using max 4 visits per site-year. Sites had to have a minimum of 2 visits within a given year and data from at least 2 years total. Models were run separately for all 17 bat species.

Fitted multi-season (dynamic) occupancy models (MacKenzie et al. 2003) to estimate trends in occupancy, colonization, and persistence while accounting for imperfect detection. Initial occupancy (ψ) was modeled as a function of elevation and water nearby, with a random effect for region. Colonization (γ) and persistence (ϕ) were modeled with time-varying intercepts and distance to road as a covariate. Detection probability (p) was modeled as a function of percent clutter, nightly mean temperature, and Juli

Elevation for each site was extracted from AWS terrain tiles (~30 m resolution, SRTM-based) using the `elevatr` package in R, distance to road and distance to harvest was extracted using the Biodiversity Pathways Human Footprint Index (not published yet). All continuous covariates were standardized by centering on their mean and dividing by their standard deviation; missing covariate values were imputed with the global mean prior to standardization.

Models were fit in a Bayesian framework using JAGS via the `jagsUI` package in R, with 30,000 iterations, 15,000 burn-in, thinning by 5, and 3 chains run in parallel, yielding 9,000 posterior samples for inference. Priors on all regression coefficients were normally distributed (mean = 0, precision = 0.368, equivalent to SD = 1.65), and the region-level random effect standard deviation received a uniform(0, 5) prior. Model convergence was assessed using Rhat diagnostics. Results are interpreted using posterior means and 95% Bayesian credible intervals. An overall occupancy trend metric (λ) was calculated as the ratio of estimated occupancy in the final year to occupancy in the initial year; values of $\lambda > 1$ with 95 CIs entirely above 1 indicate an increase in occupancy, values $\lambda < 1$ with 95 CIs entirely below 1 indicate a decline, and values with CIs overlapping 1 indicate uncertain or stable trends.

4 Results

In 2025 a total of 6 grid sites were sampled in SE Alaska and 3 newly established sites were sampled around the Anchorage area (Figure 2).

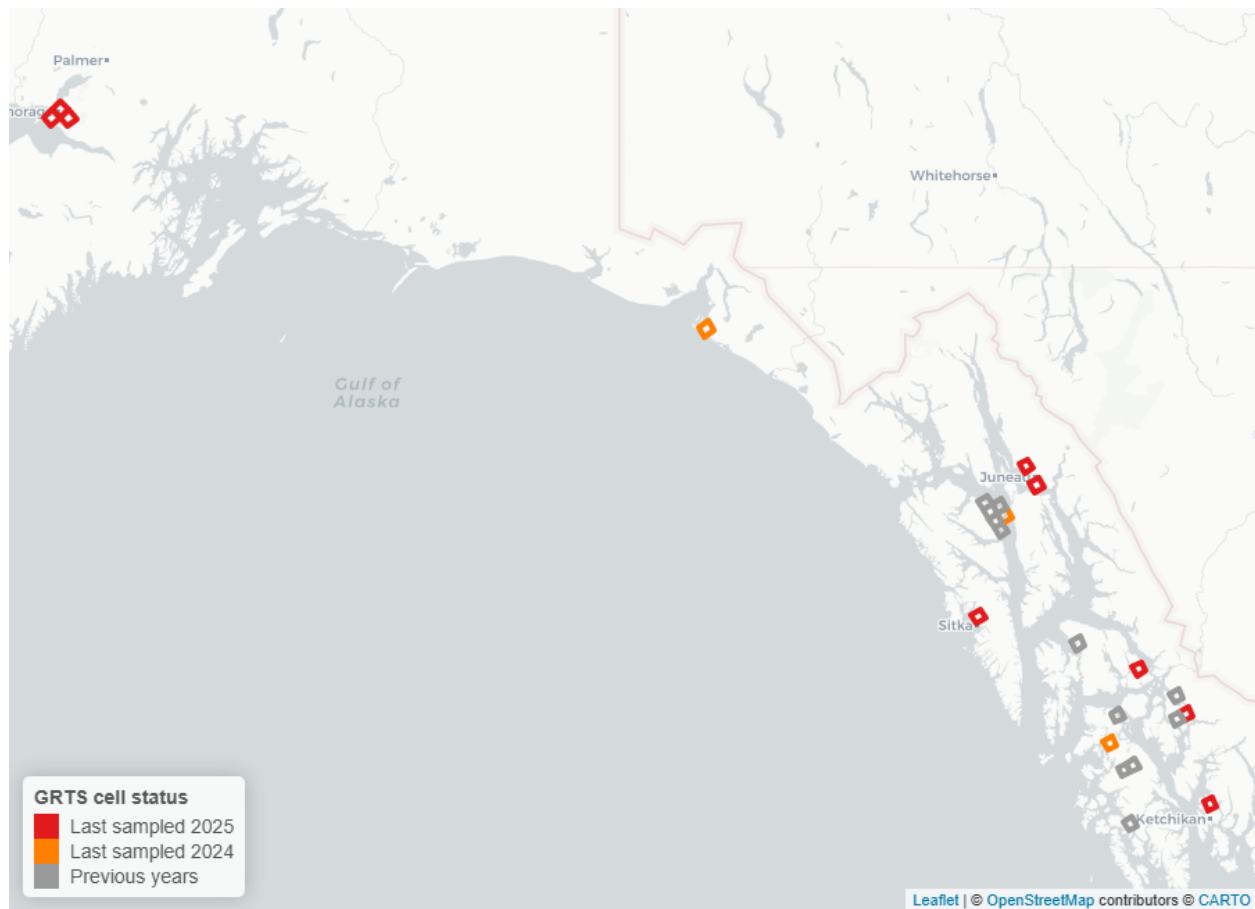


Figure 2: Acoustic bat monitoring sites in Alaska in 2025. NABat grid cells monitored (with two or more stationary sites) in 2025 appear in red and those monitored in 2024 in orange. Cells monitored in previous years but not in 2024 or 2025 appear in grey.

4.1 Activity Trends

Table 2: Provincial stationary and transect trend estimates from 2017-2024 NABat monitoring in BC adjusted for covariates. Trends estimated on a logarithmic scale, thus a trend estimate of 0.05 corresponds to an annual change of 5%. Significant results (p value < 0.05) are highlighted in bold.

Species	Stationary Models	
	Trend Estimate $\pm$ 95%CI	P-value
Hoary Bat	-0.31 $\pm$ 0.36	0.094
Silver-haired Bat	0.37 $\pm$ 0.15	<0.001
California Myotis	0.18 $\pm$ 0.12	0.004
Long-eared Myotis	0.05 $\pm$ 0.11	0.364
Little Brown Myotis	0.13 $\pm$ 0.13	0.042
Long-legged Myotis		
Yuma Myotis		

4.2 Dynamic Occupancy

Temperature was the most consistent detection covariate—warmer nights equate to higher detection for most species. There were some clutter effects on detection probability that were species-specific. Julian date effects were mixed across species.

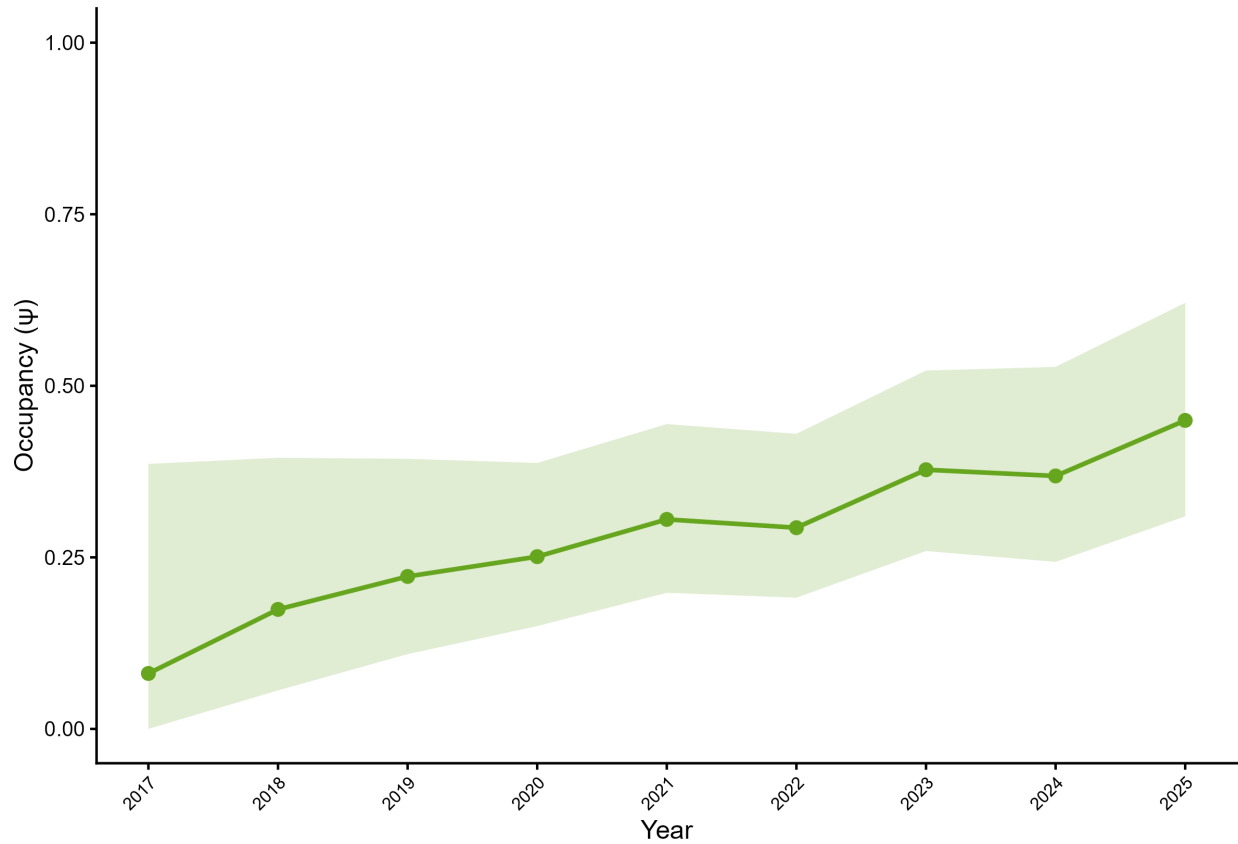
Species specific results can be found on [Appendix A](#)

5 Appendix A

5.1 Hoary Bat (*Lasiurus cinereus*)

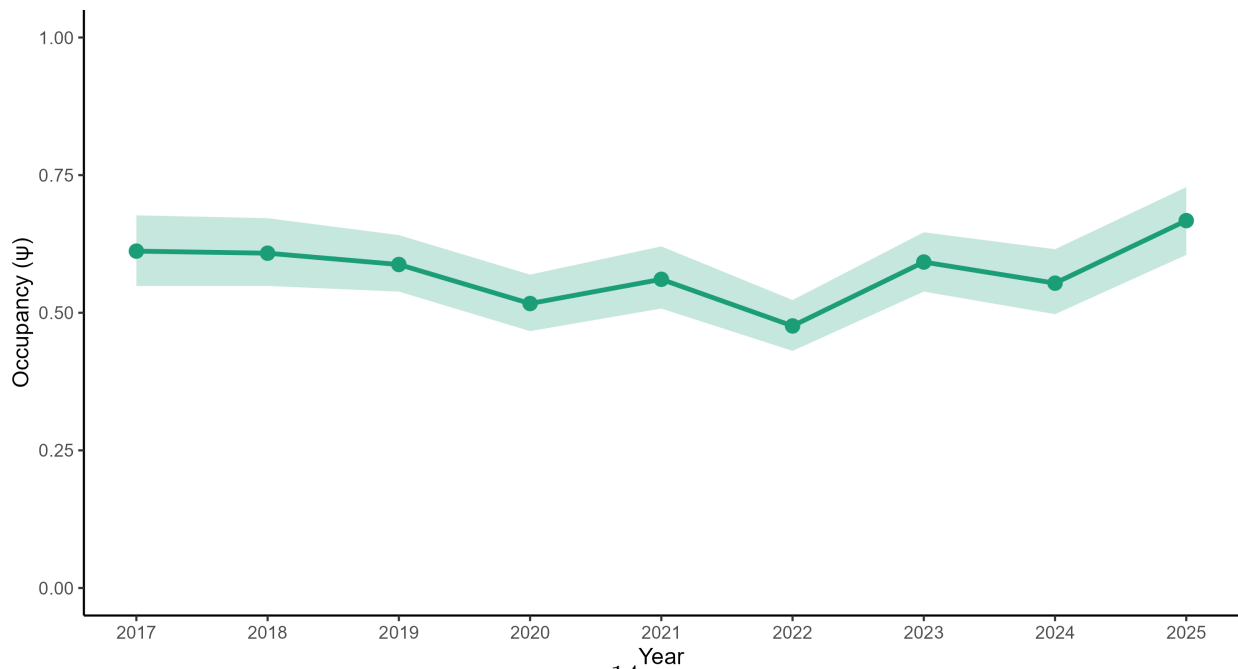
LACI — Alaska Occupancy (2017–2025)

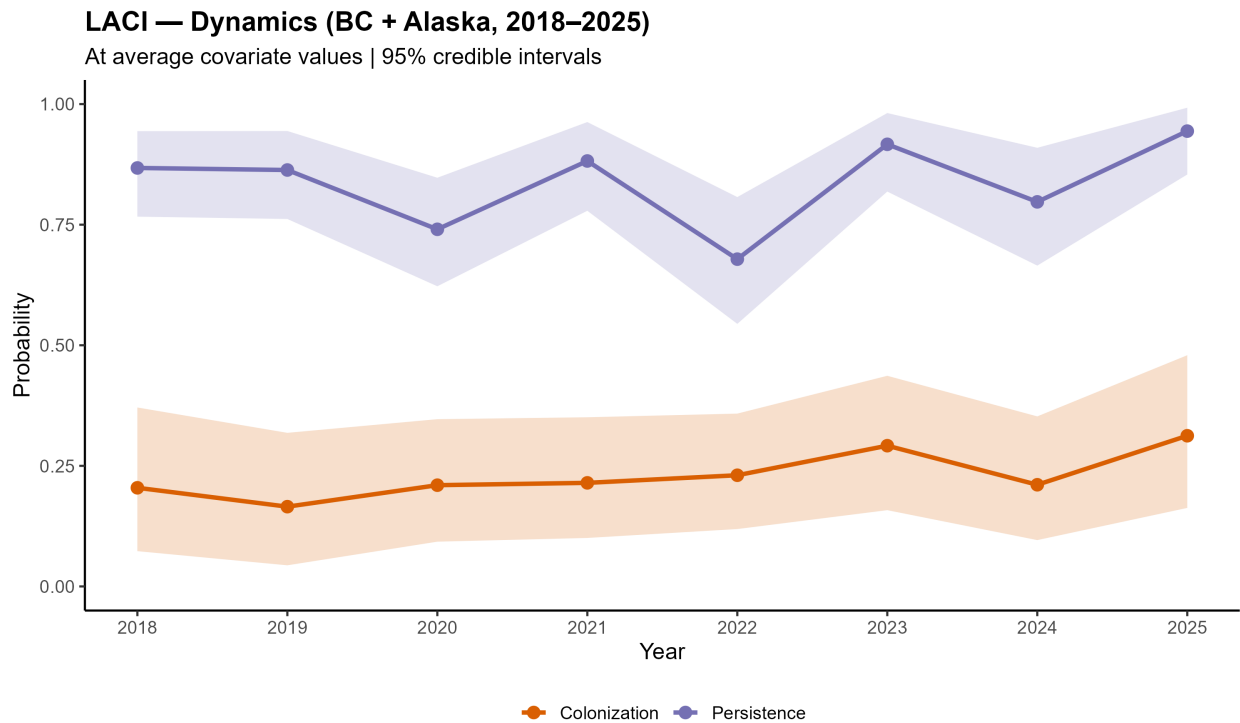
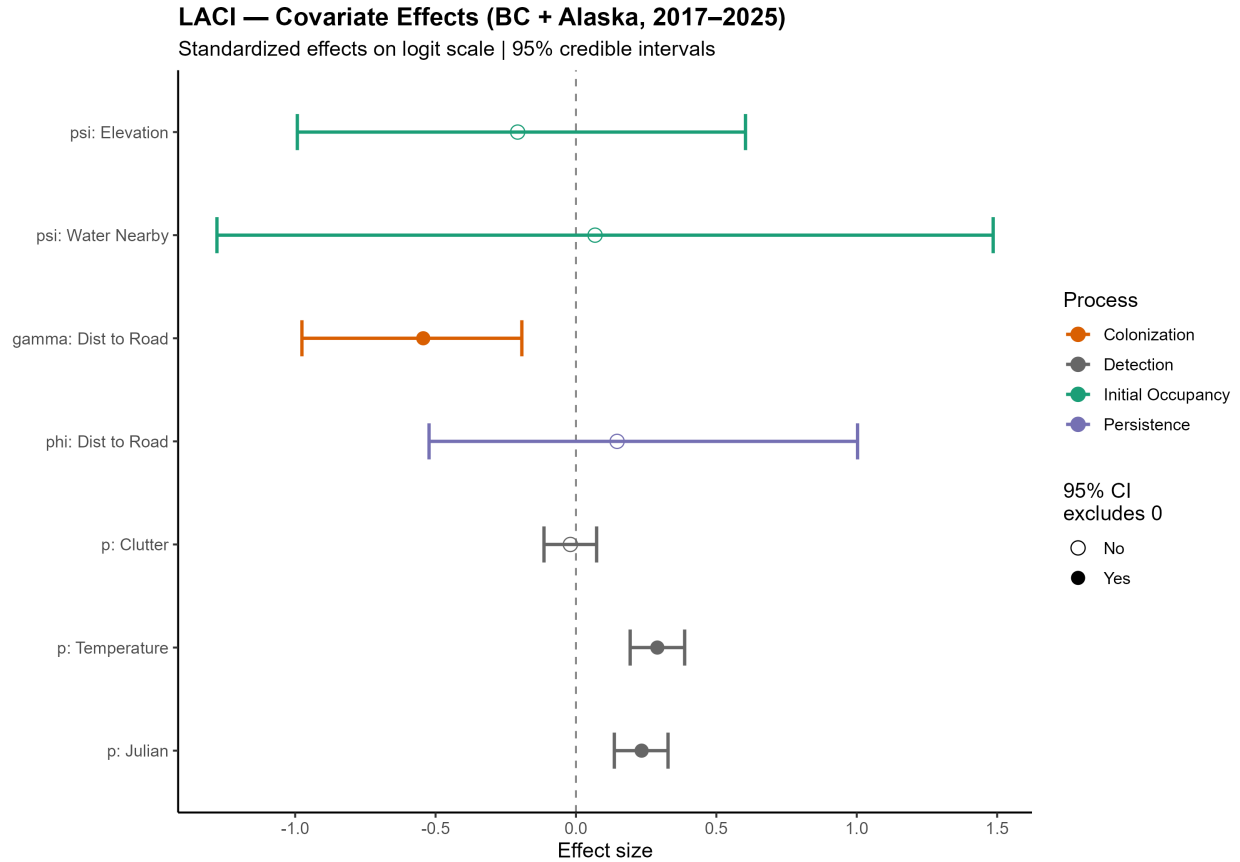
Model-expected occupancy reconstructed from parameters | 95% credible intervals | $n = 14$



LACI — Occupancy Trend (BC + Alaska, 2017–2025)

$\lambda = 1.09$ (0.95, 1.26) | Mean $p = 0.54$

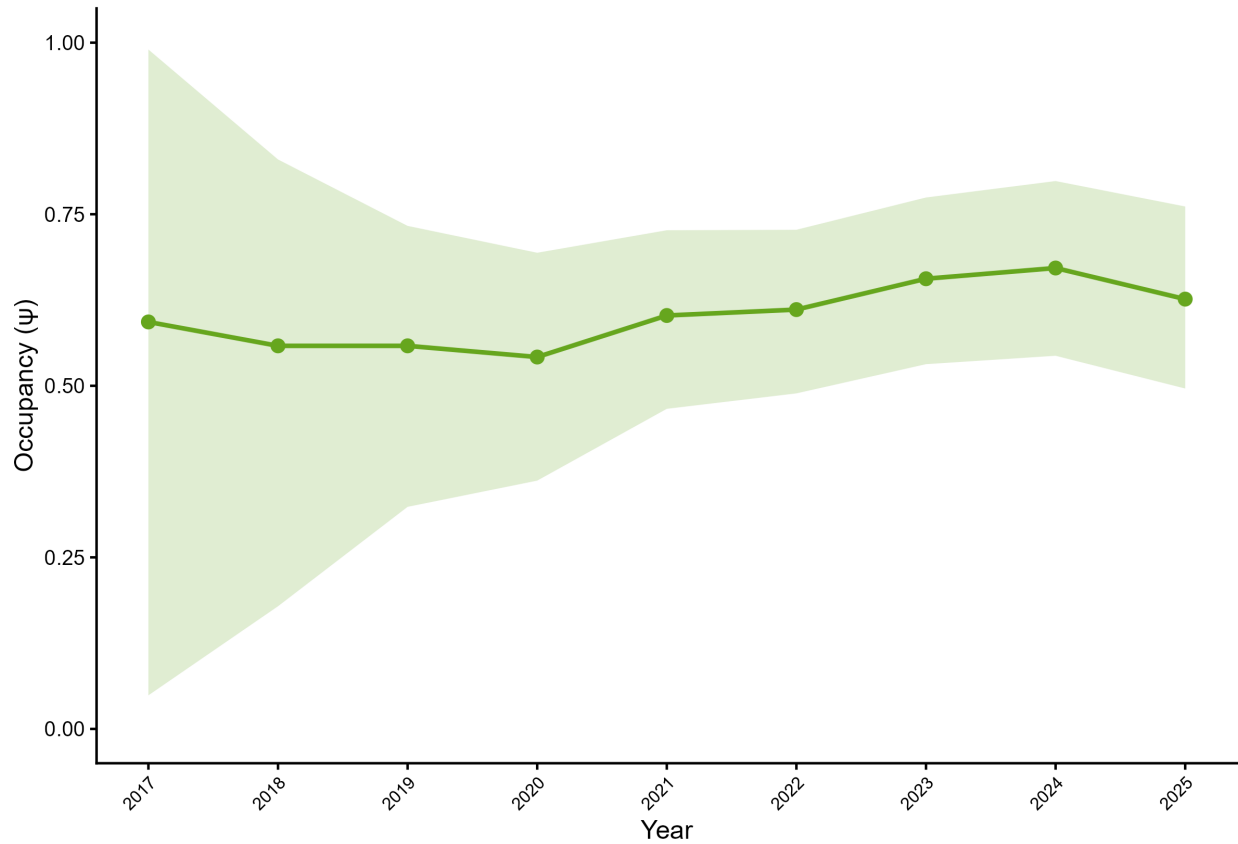




##Silver-haired Bat (*Lasiyonicteris noctivagans*)

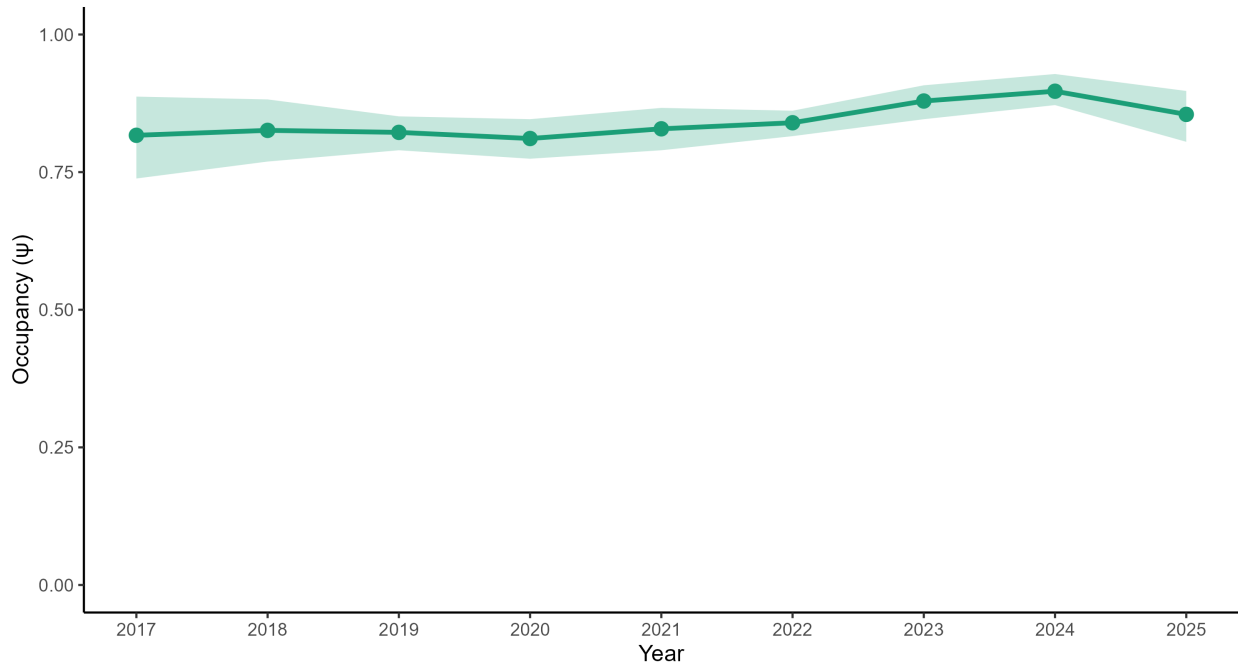
LANO — Alaska Occupancy (2017–2025)

Model-expected occupancy reconstructed from parameters | 95% credible intervals | $n = 14$



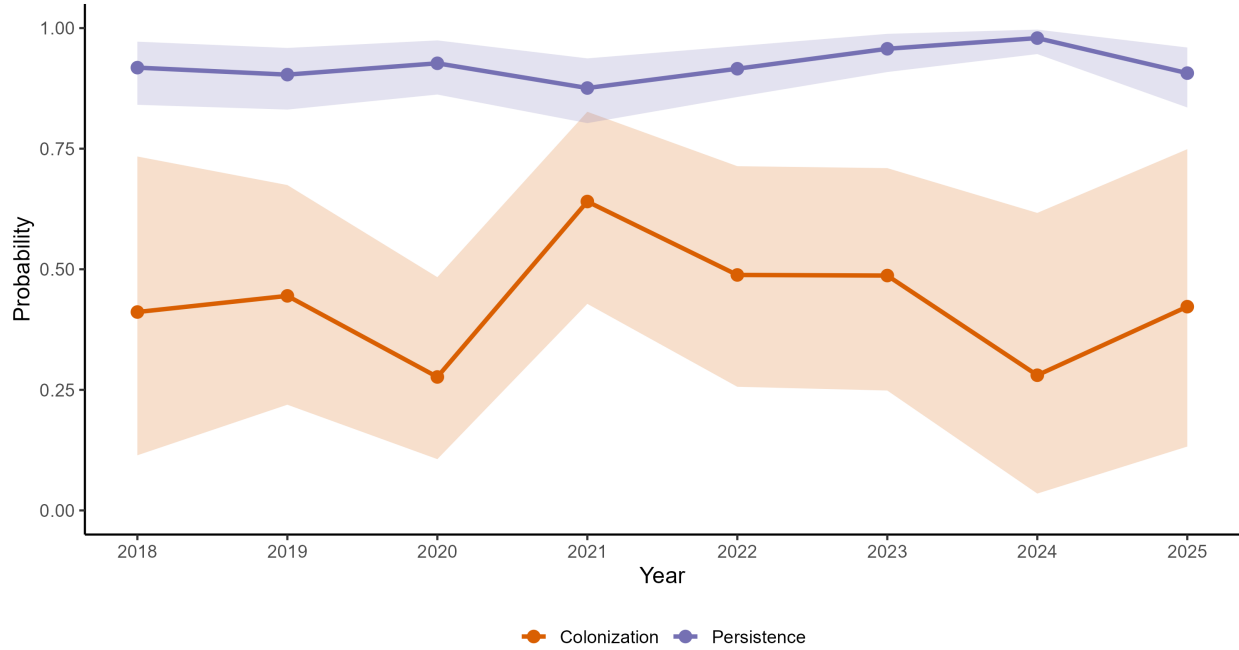
LANO — Occupancy Trend (BC + Alaska, 2017–2025)

$\lambda = 1.05$ (0.95, 1.17) | Mean $p = 0.77$



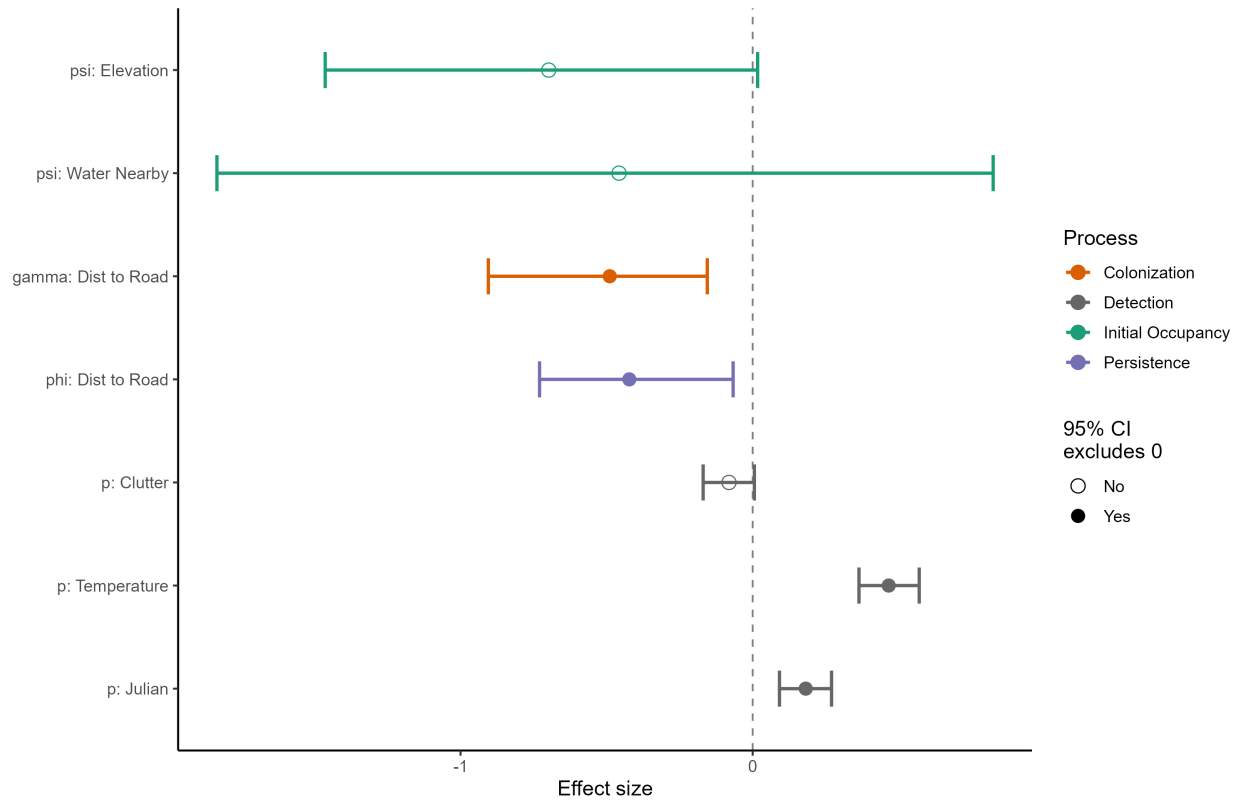
LANO — Dynamics (BC + Alaska, 2018–2025)

At average covariate values | 95% credible intervals



LANO — Covariate Effects (BC + Alaska, 2017–2025)

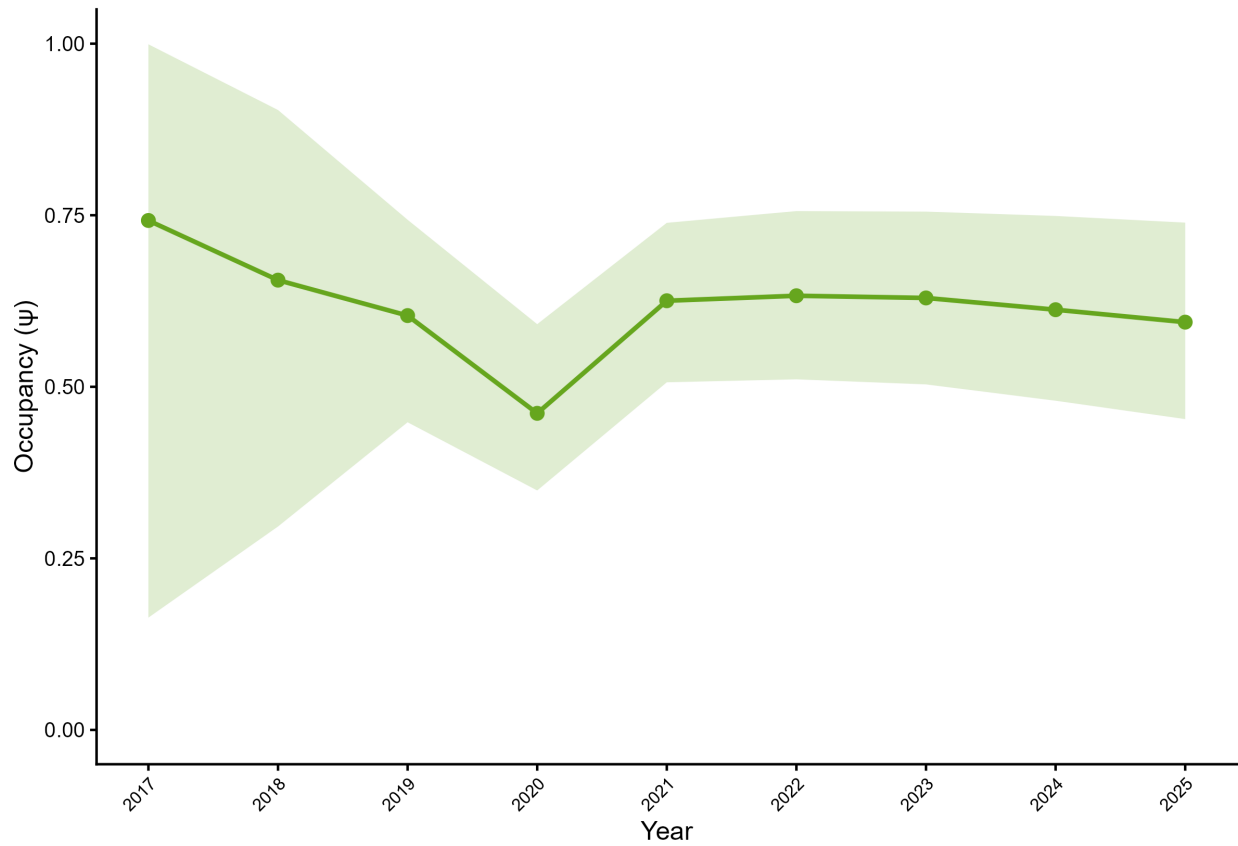
Standardized effects on logit scale | 95% credible intervals



5.2 California Myotis (*Myotis californicus*)

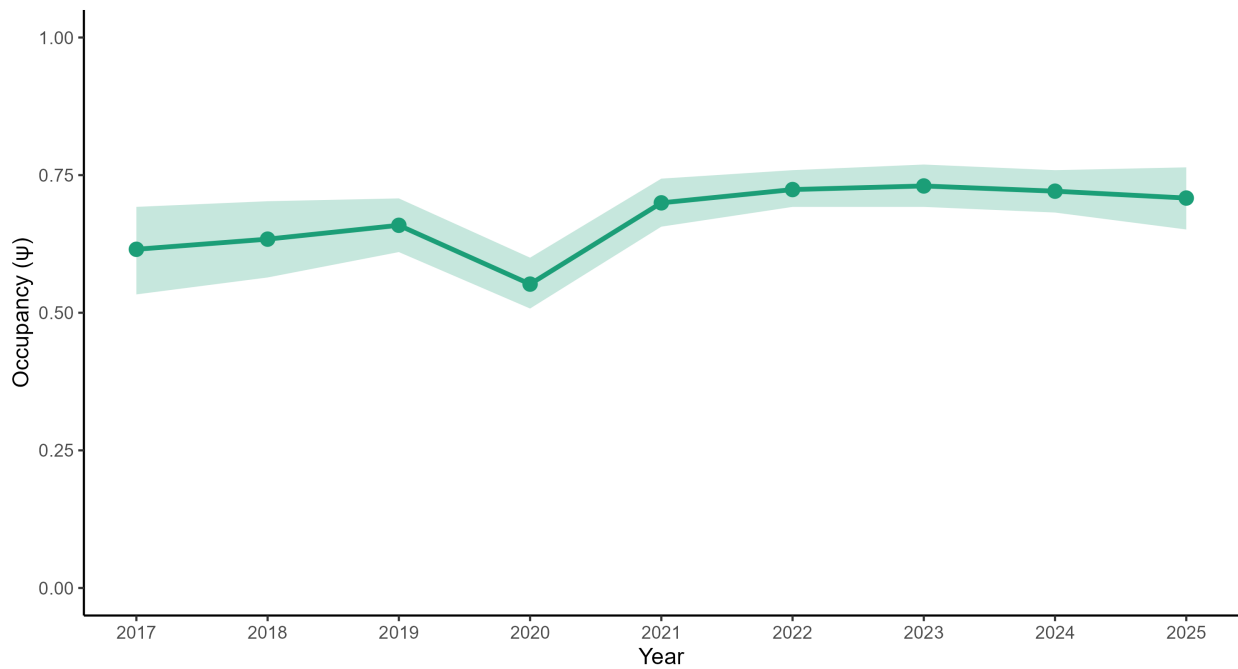
MYCA — Alaska Occupancy (2017–2025)

Model-expected occupancy reconstructed from parameters | 95% credible intervals | $n = 14$



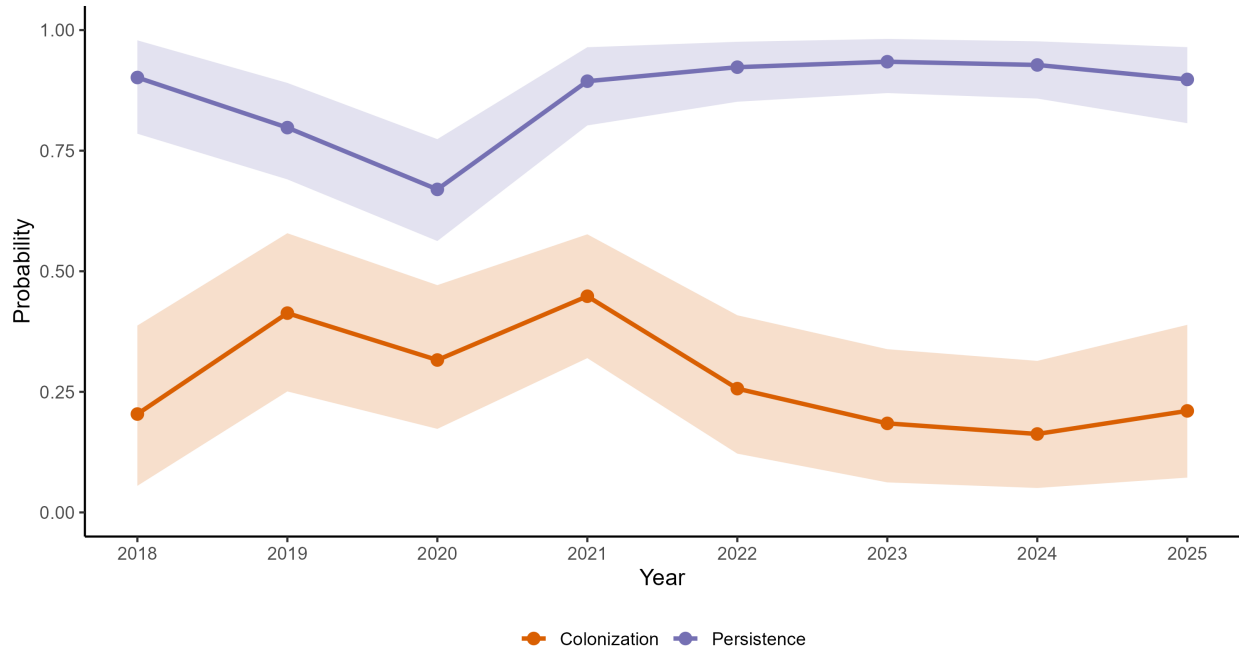
MYCA — Occupancy Trend (BC + Alaska, 2017–2025)

$\lambda = 1.16$ (1.00, 1.35) | Mean $p = 0.65$



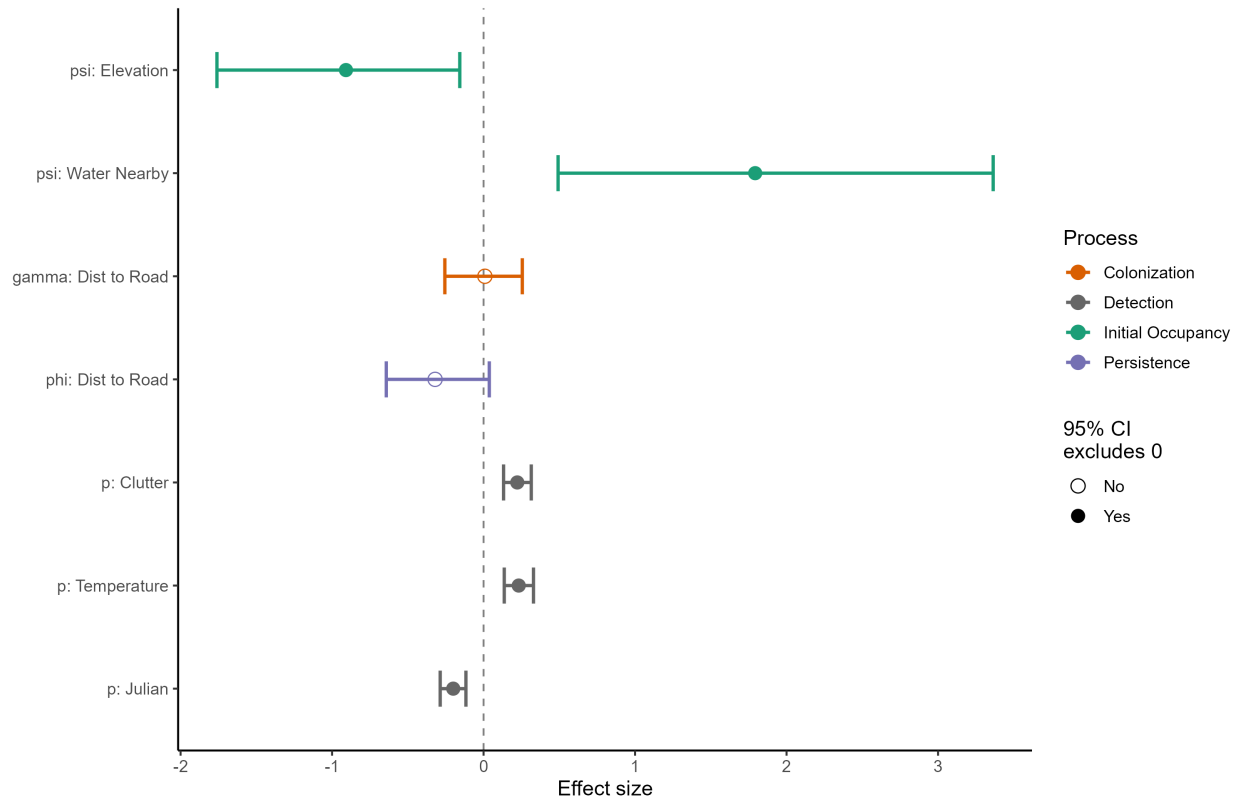
MYCA — Dynamics (BC + Alaska, 2018–2025)

At average covariate values | 95% credible intervals



MYCA — Covariate Effects (BC + Alaska, 2017–2025)

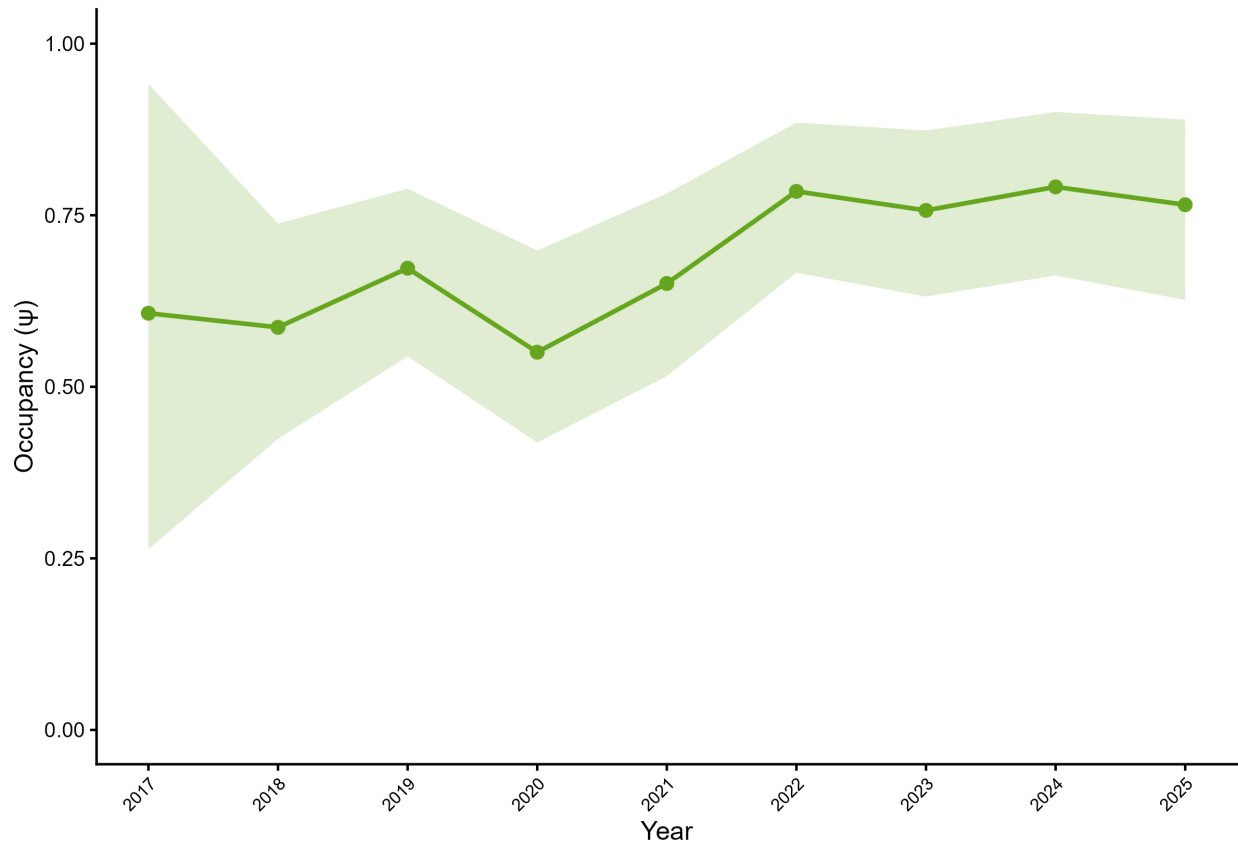
Standardized effects on logit scale | 95% credible intervals



5.3 Long-eared Myotis (*Myotis evotis*)

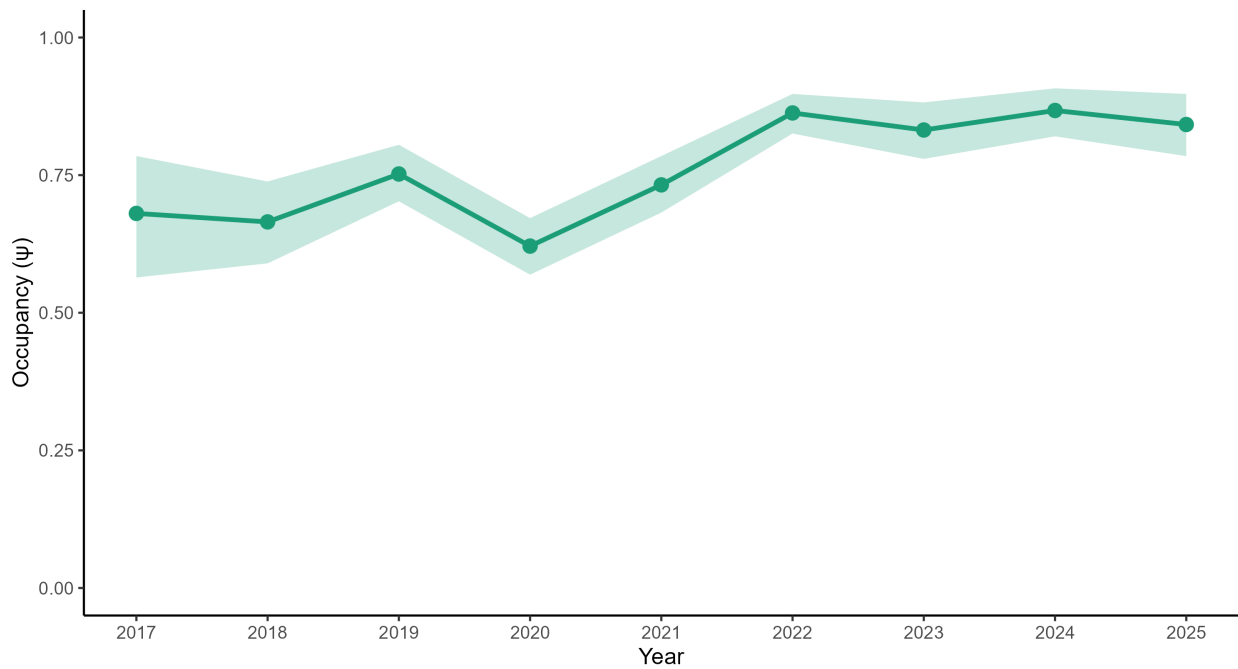
MYEV — Alaska Occupancy (2017–2025)

Model-expected occupancy reconstructed from parameters | 95% credible intervals | $n = 14$



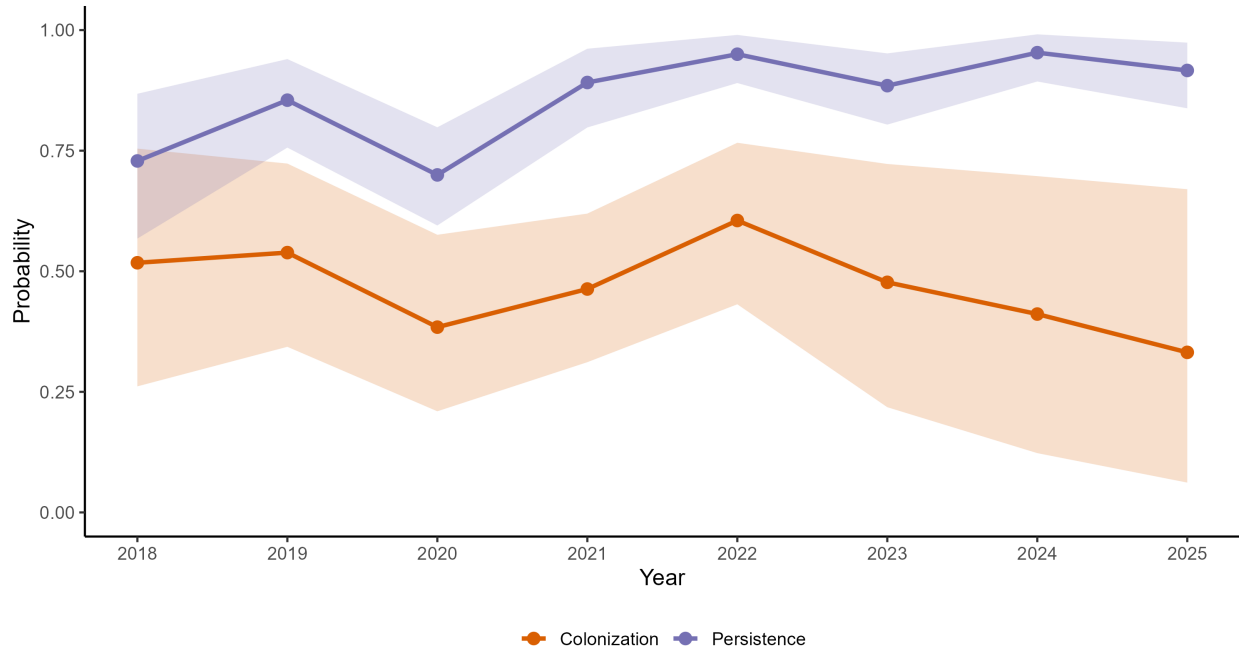
MYEV — Occupancy Trend (BC + Alaska, 2017–2025)

$\lambda = 1.25$ (1.05, 1.50) | Mean $p = 0.58$



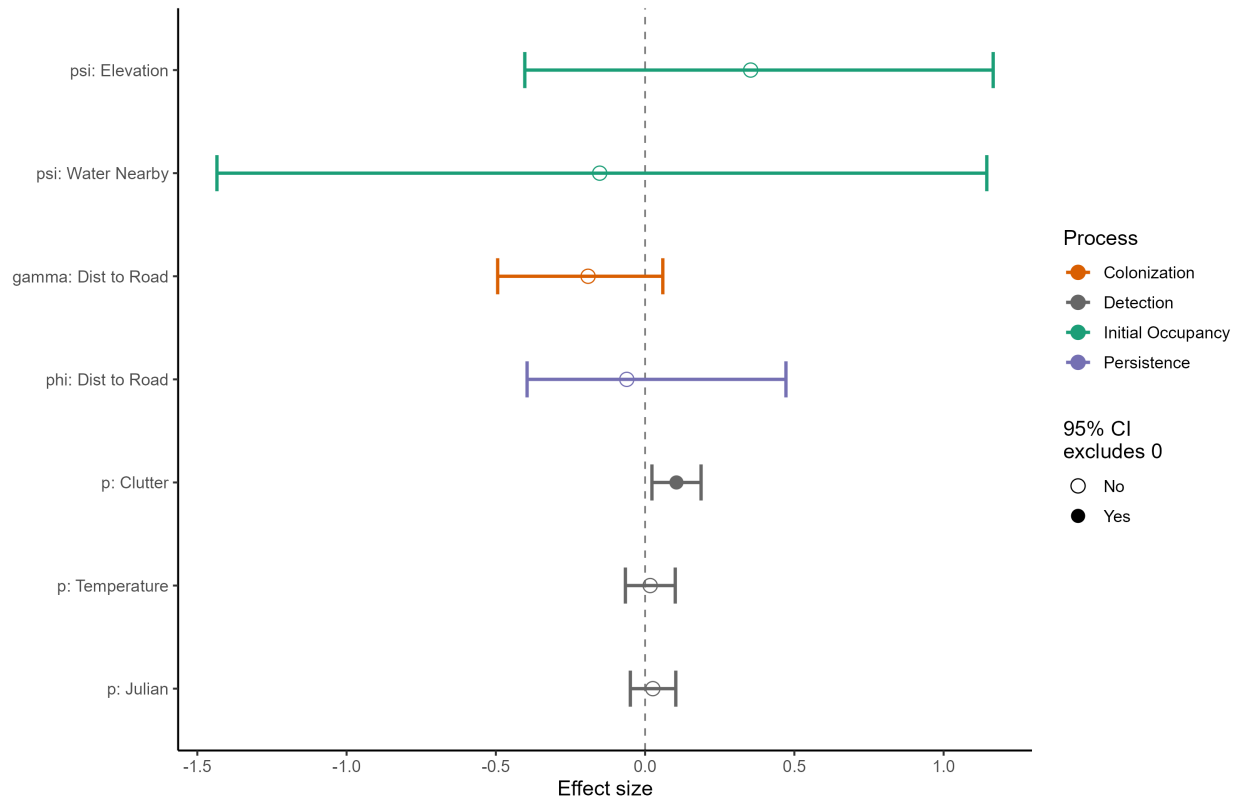
MYEV — Dynamics (BC + Alaska, 2018–2025)

At average covariate values | 95% credible intervals



MYEV — Covariate Effects (BC + Alaska, 2017–2025)

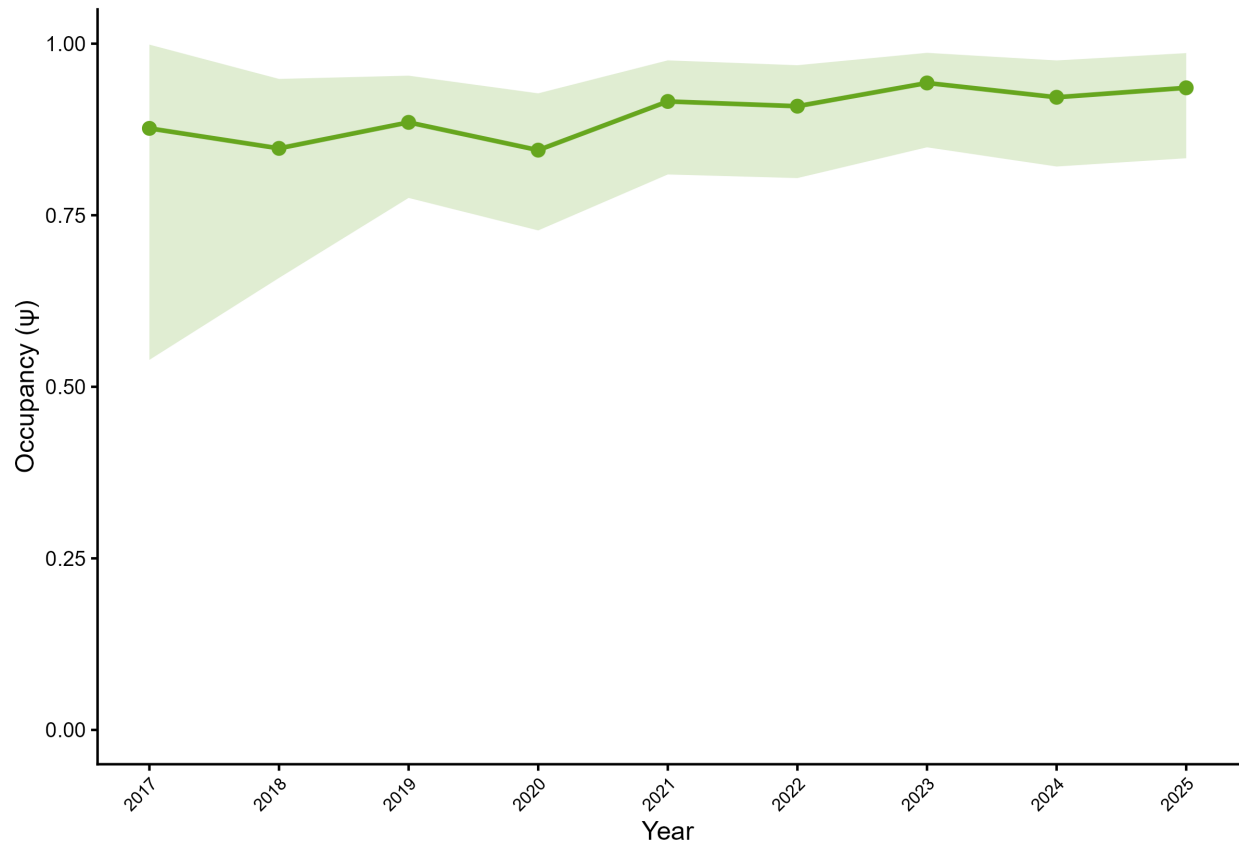
Standardized effects on logit scale | 95% credible intervals



5.4 Little Brown Myotis (*Myotis lucifugus*)

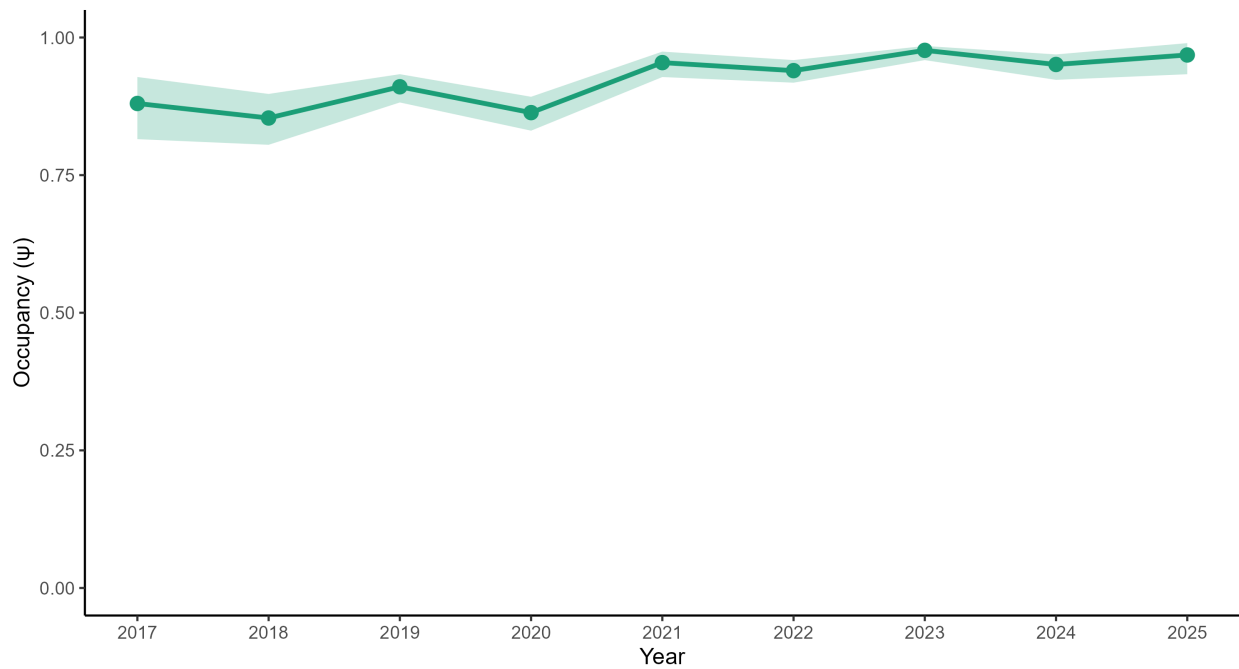
MYLU — Alaska Occupancy (2017–2025)

Model-expected occupancy reconstructed from parameters | 95% credible intervals | $n = 14$



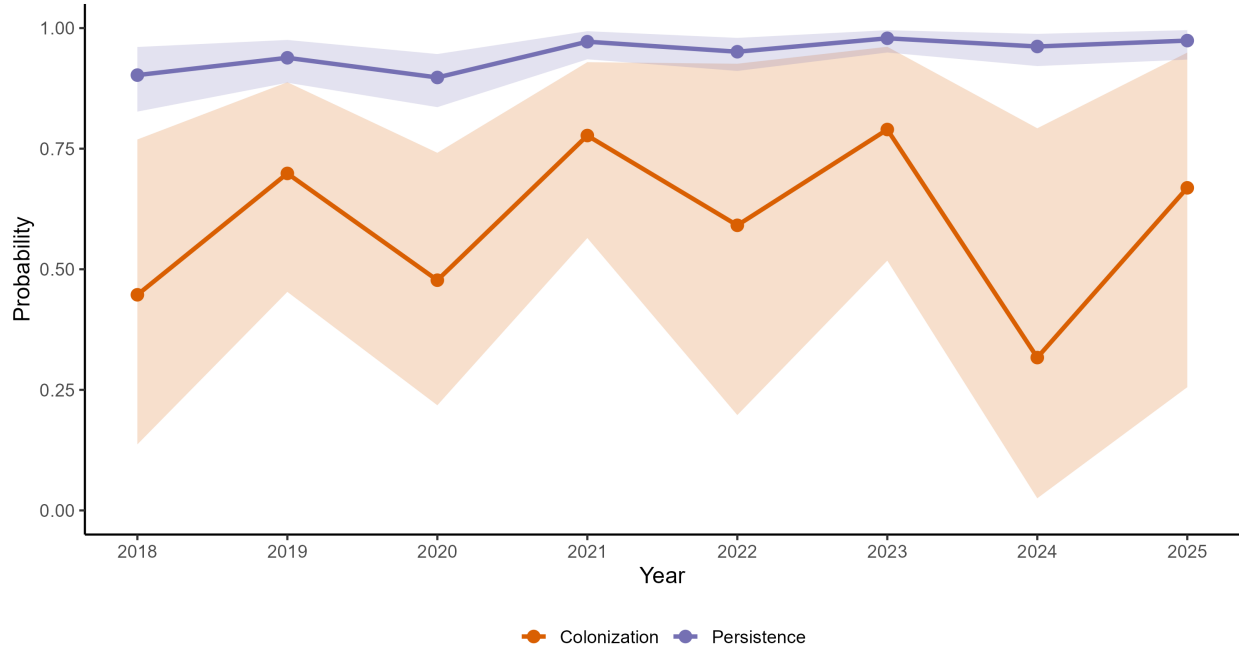
MYLU — Occupancy Trend (BC + Alaska, 2017–2025)

$\lambda = 1.10$ (1.03, 1.19) | Mean $p = 0.83$



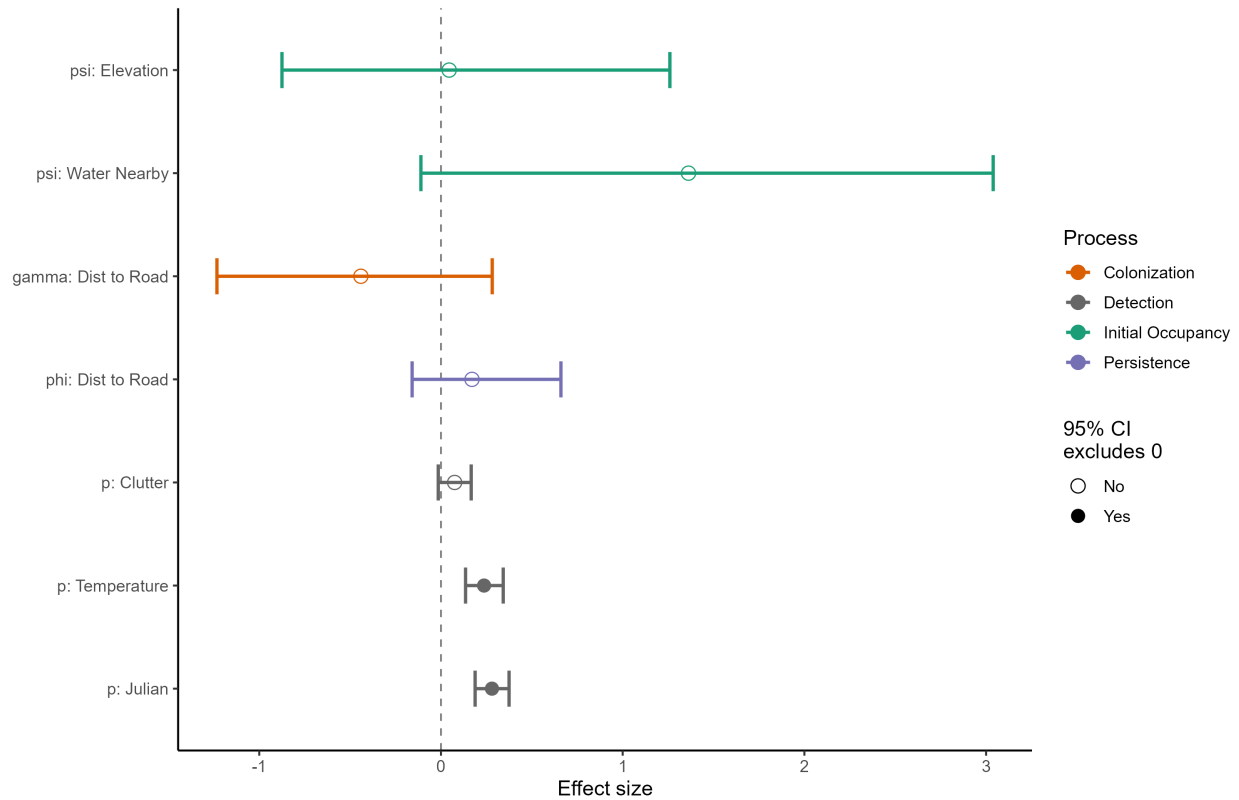
MYLU — Dynamics (BC + Alaska, 2018–2025)

At average covariate values | 95% credible intervals



MYLU — Covariate Effects (BC + Alaska, 2017–2025)

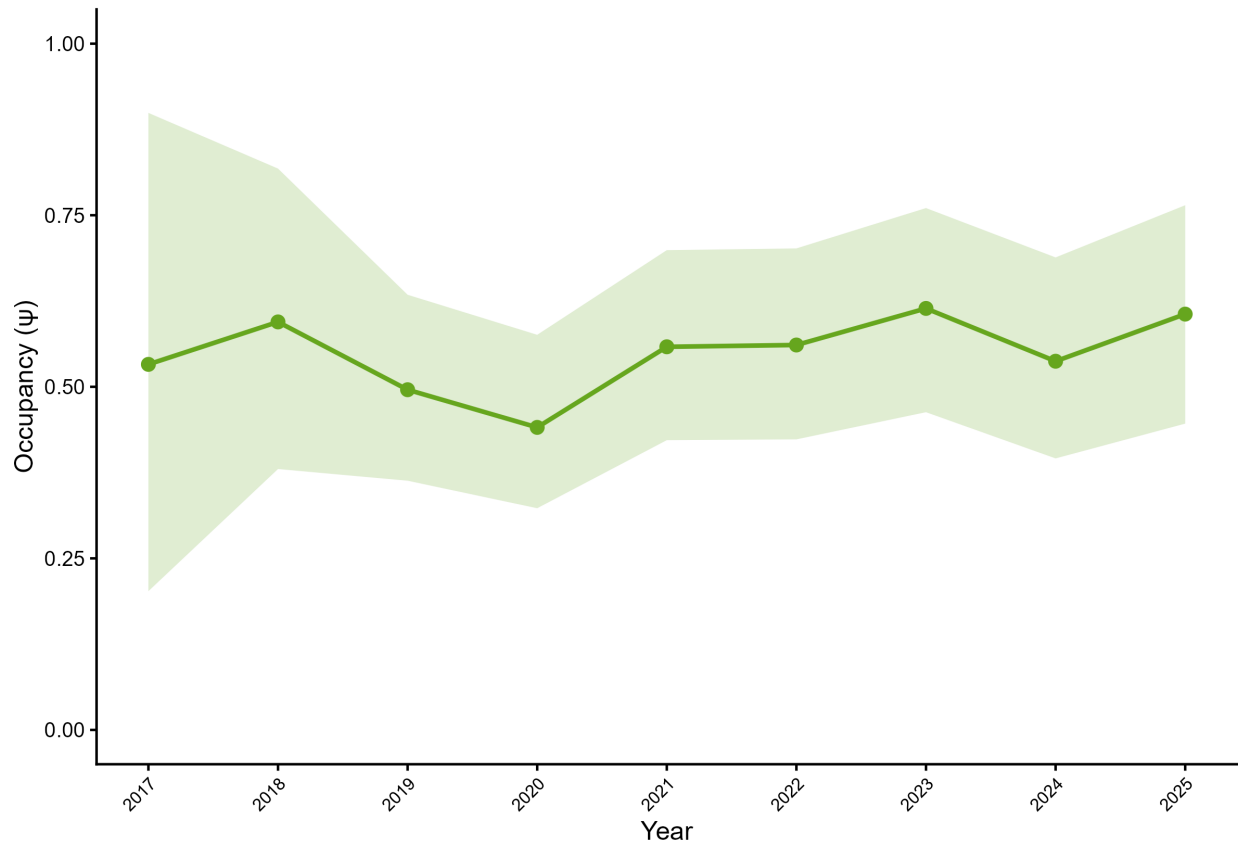
Standardized effects on logit scale | 95% credible intervals



5.5 Long-legged Myotis (*Myotis volans*)

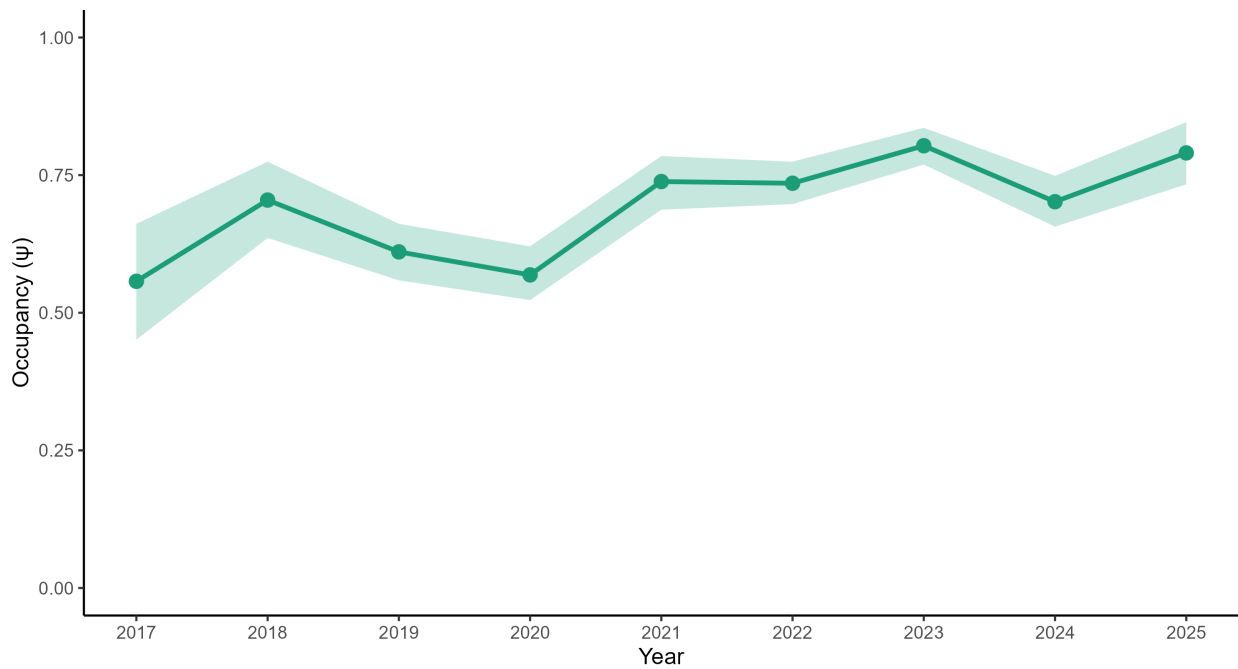
MYVO — Alaska Occupancy (2017–2025)

Model-expected occupancy reconstructed from parameters | 95% credible intervals | $n = 14$



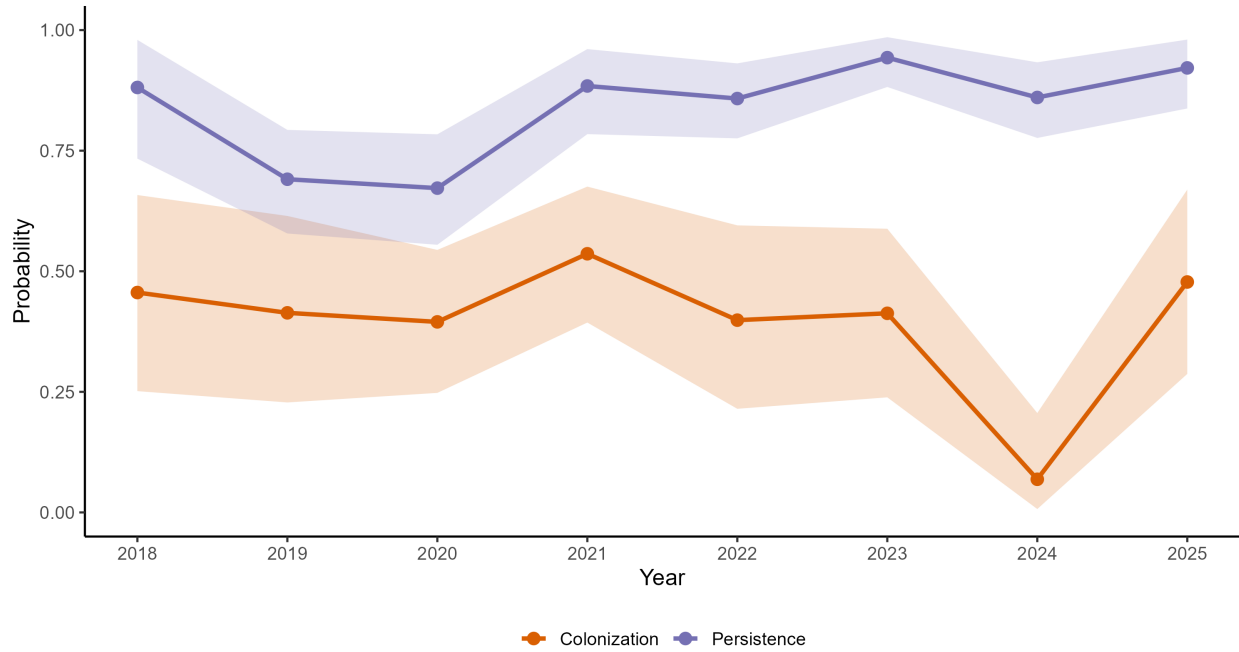
MYVO — Occupancy Trend (BC + Alaska, 2017–2025)

$\lambda = 1.43$ (1.18, 1.77) | Mean $p = 0.61$



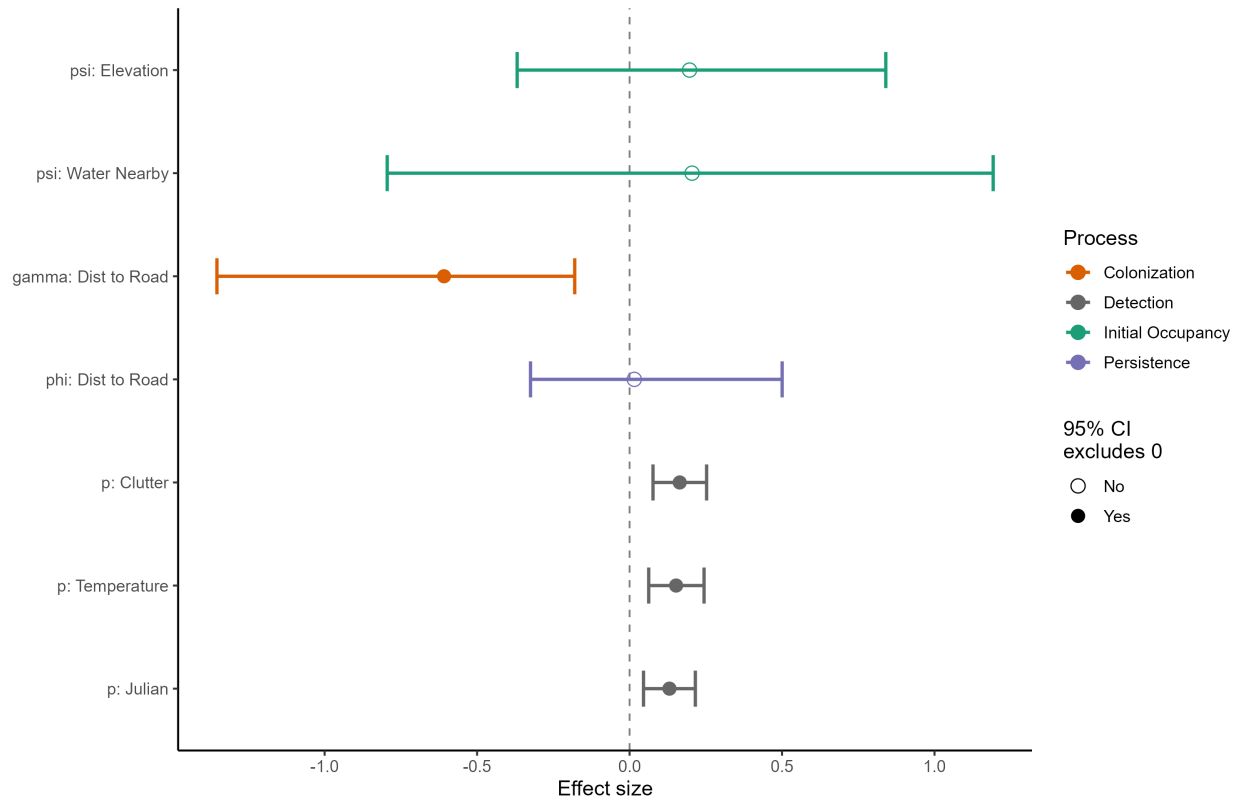
MYVO — Dynamics (BC + Alaska, 2018–2025)

At average covariate values | 95% credible intervals



MYVO — Covariate Effects (BC + Alaska, 2017–2025)

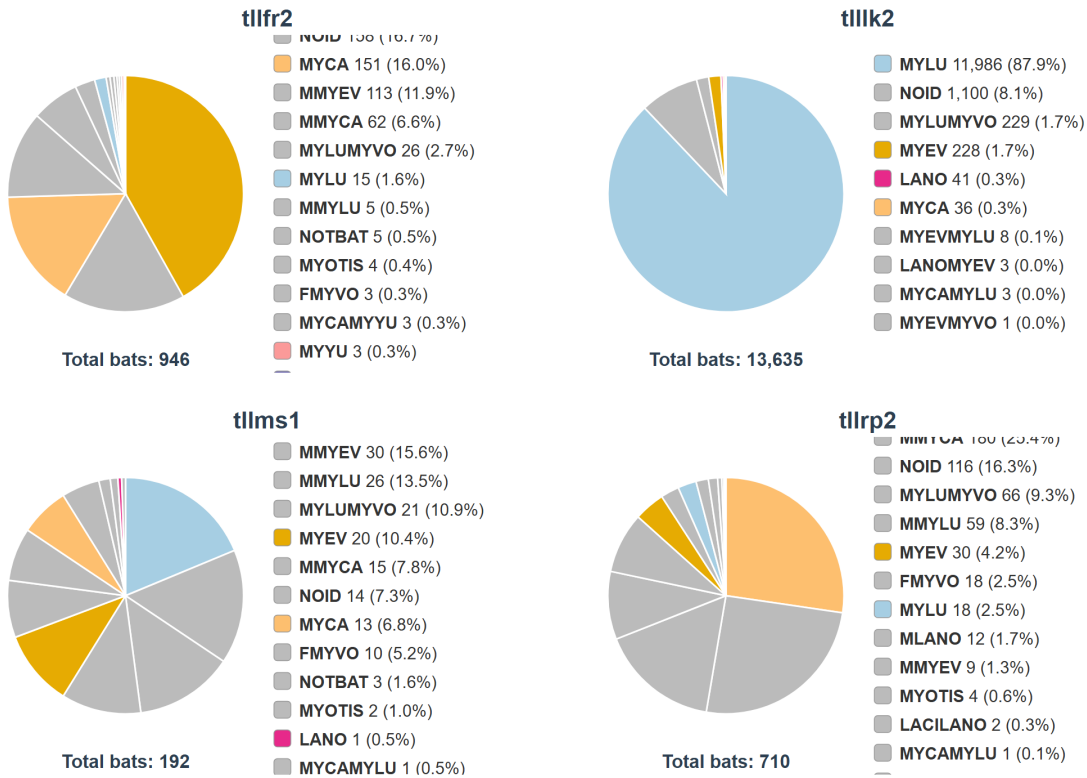
Standardized effects on logit scale | 95% credible intervals



6 Appendix B

2025 Species detection summaries by GRTS cell.

Grid cell 5006



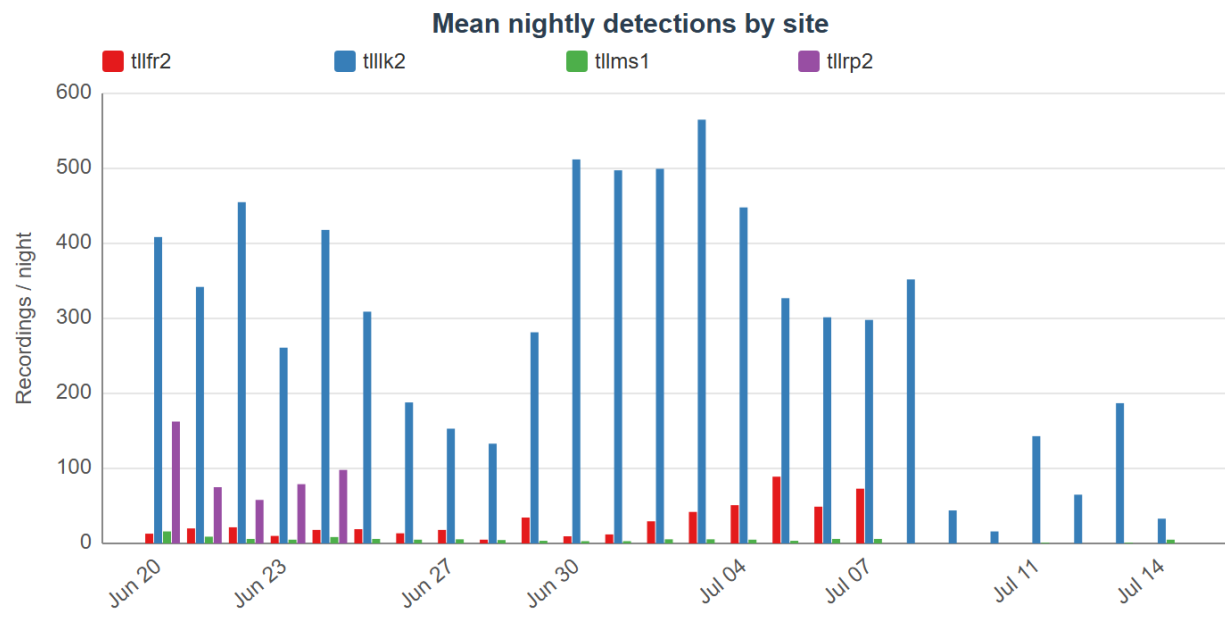
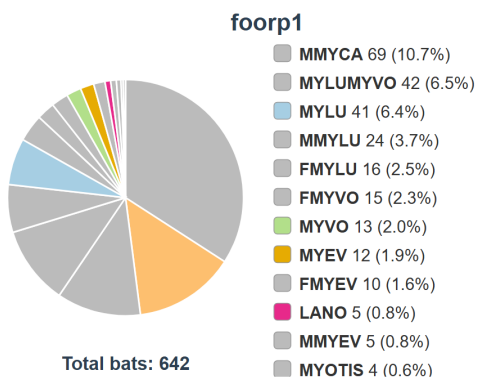
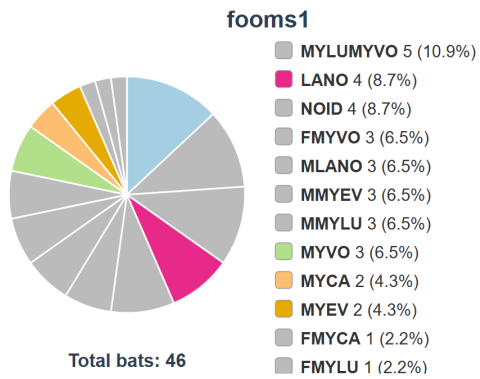
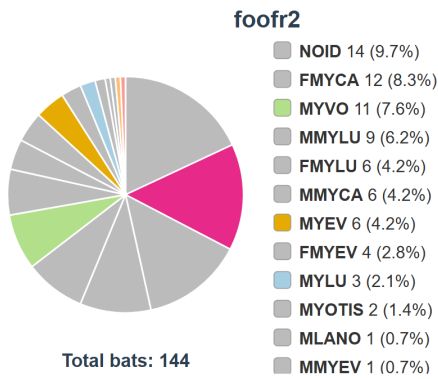


Figure 3: 2025 Nightly activity by site, cell 5006



Figure 4: 2025 Detection times by species and site, cell 5006

Grid cell 18318



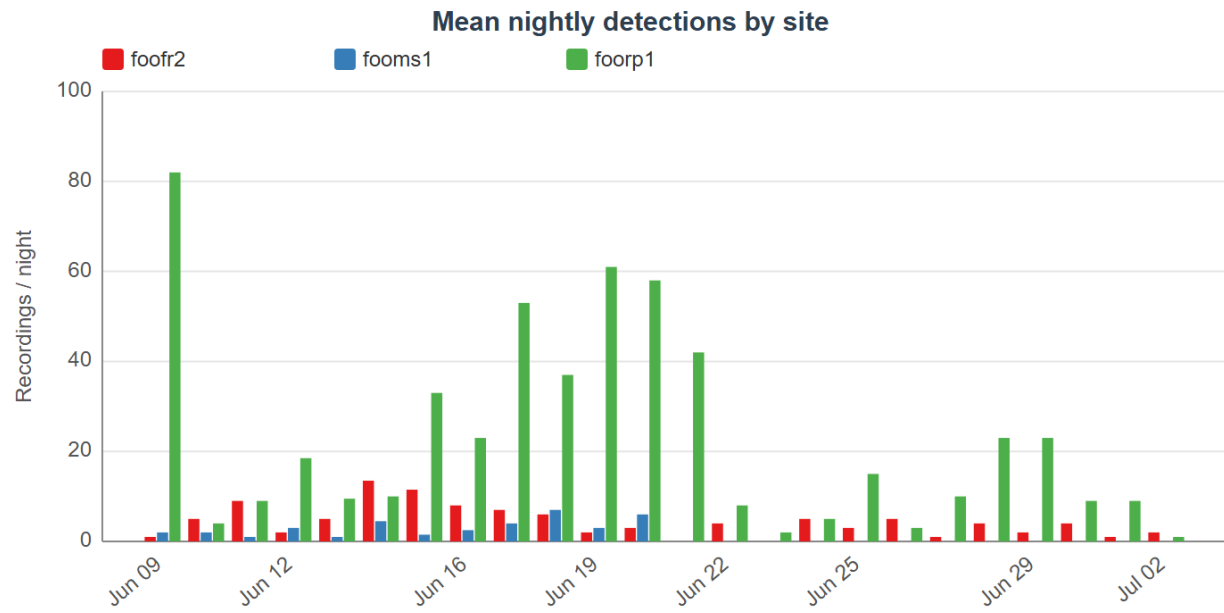


Figure 5: 2025 Nightly activity by site, cell 18318

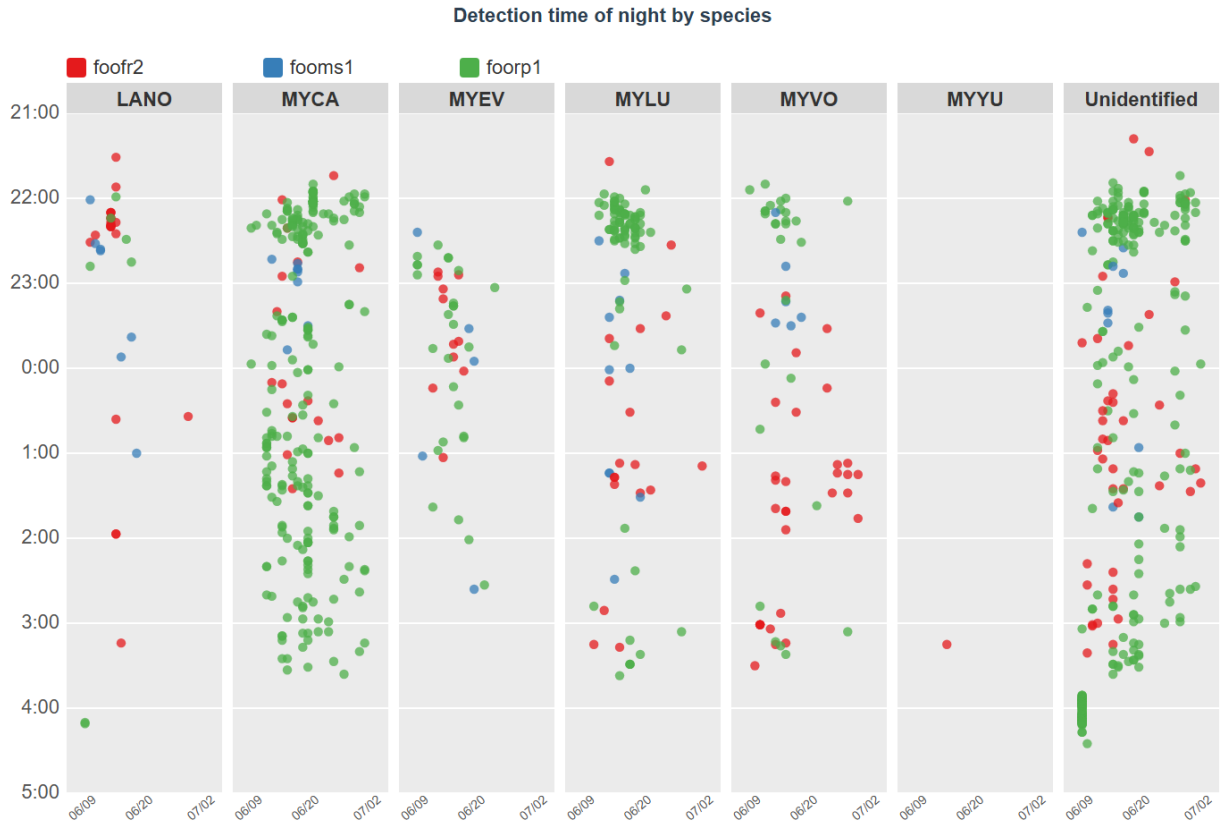
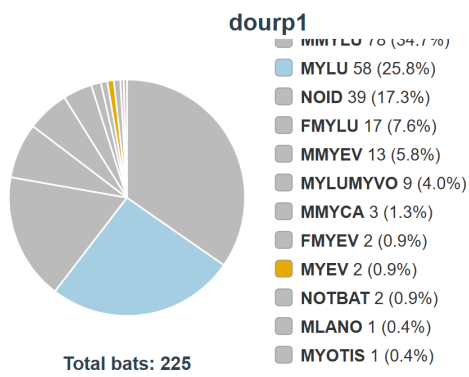
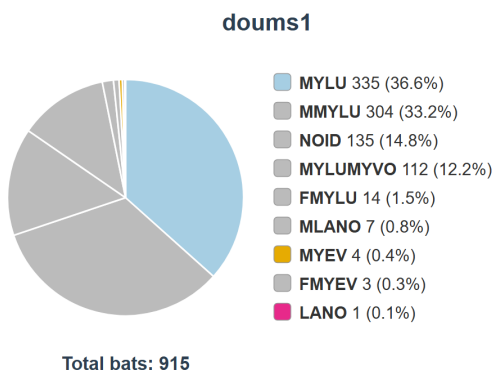
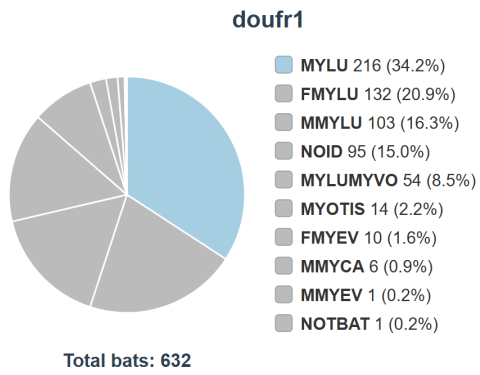
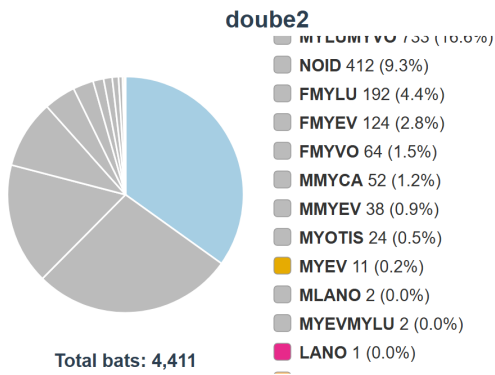


Figure 6: 2025 Detection times by species and site, cell 18318

Grid cell 31438



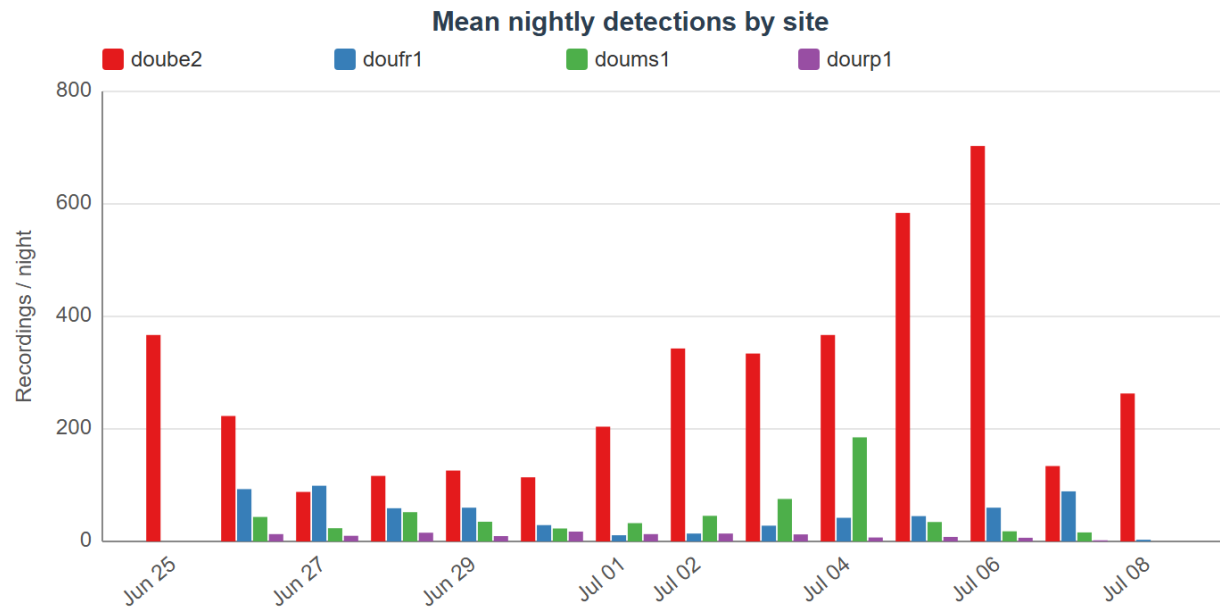


Figure 7: 2025 Nightly activity by site, cell 31438

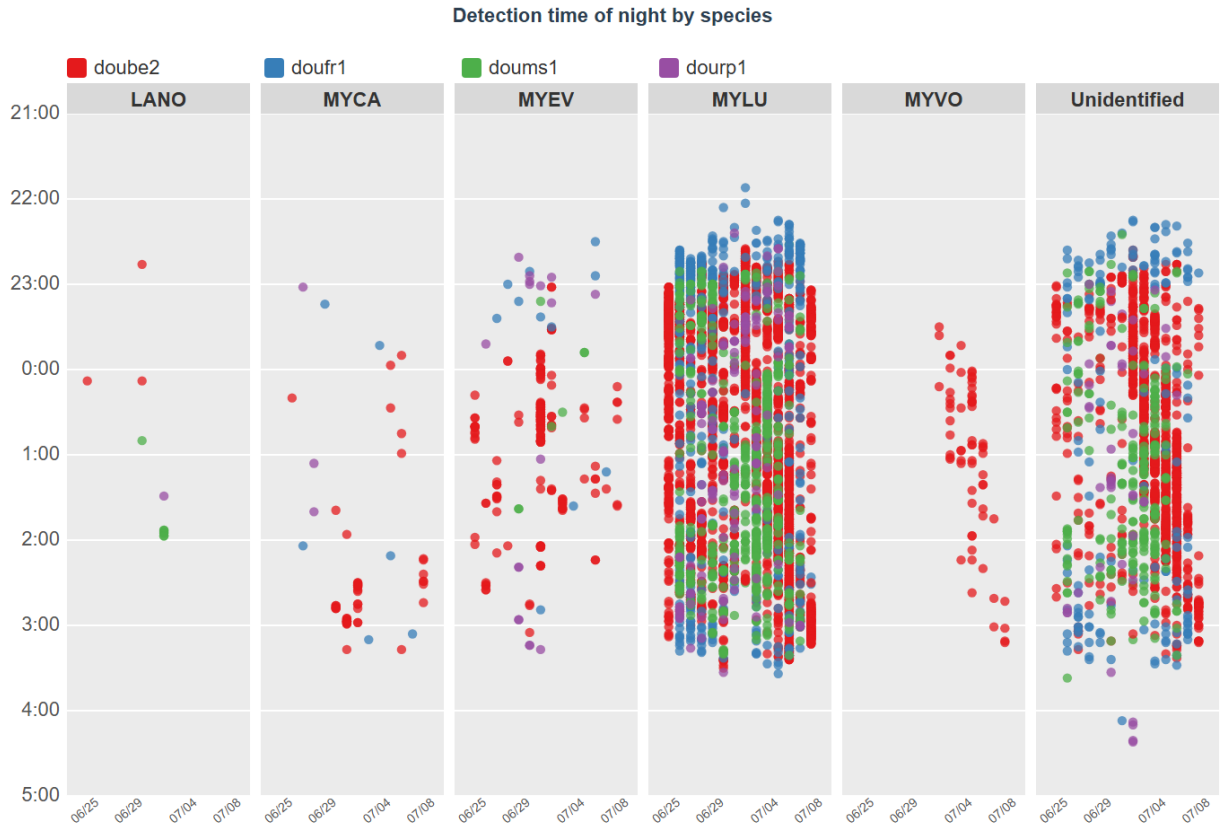
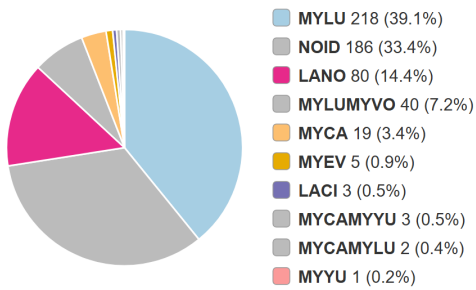


Figure 8: 2025 Detection times by species and site, cell 31438

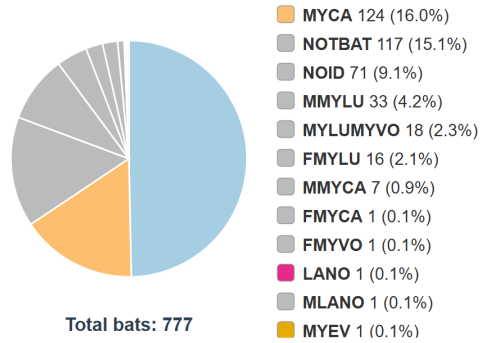
Grid cell 33486

stkbe3



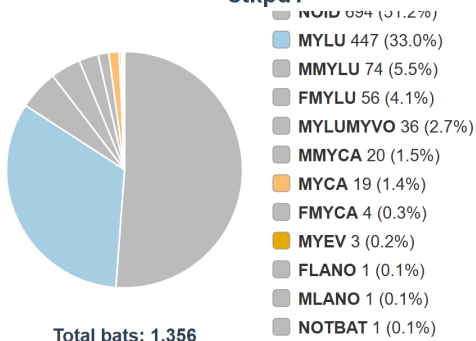
Total bats: 557

stkms2



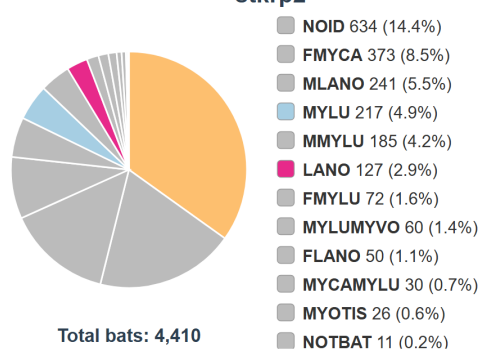
Total bats: 777

stkpdp1



Total bats: 1,356

stkrp2



Total bats: 4,410

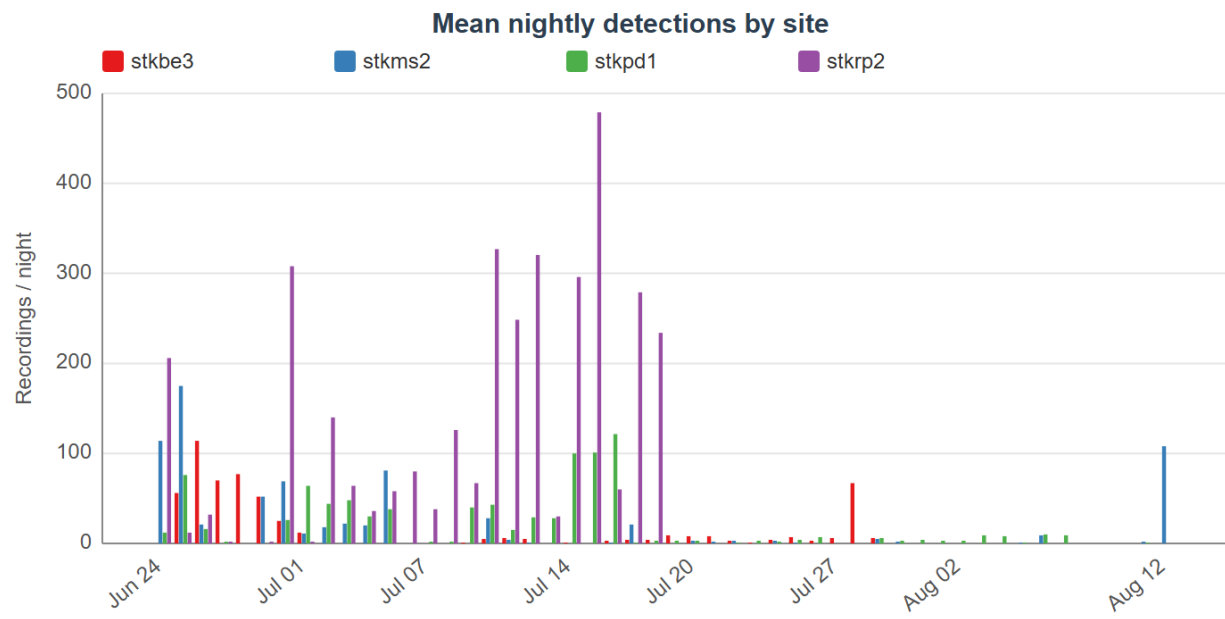


Figure 9: 2025 Nightly activity by site, cell 33486

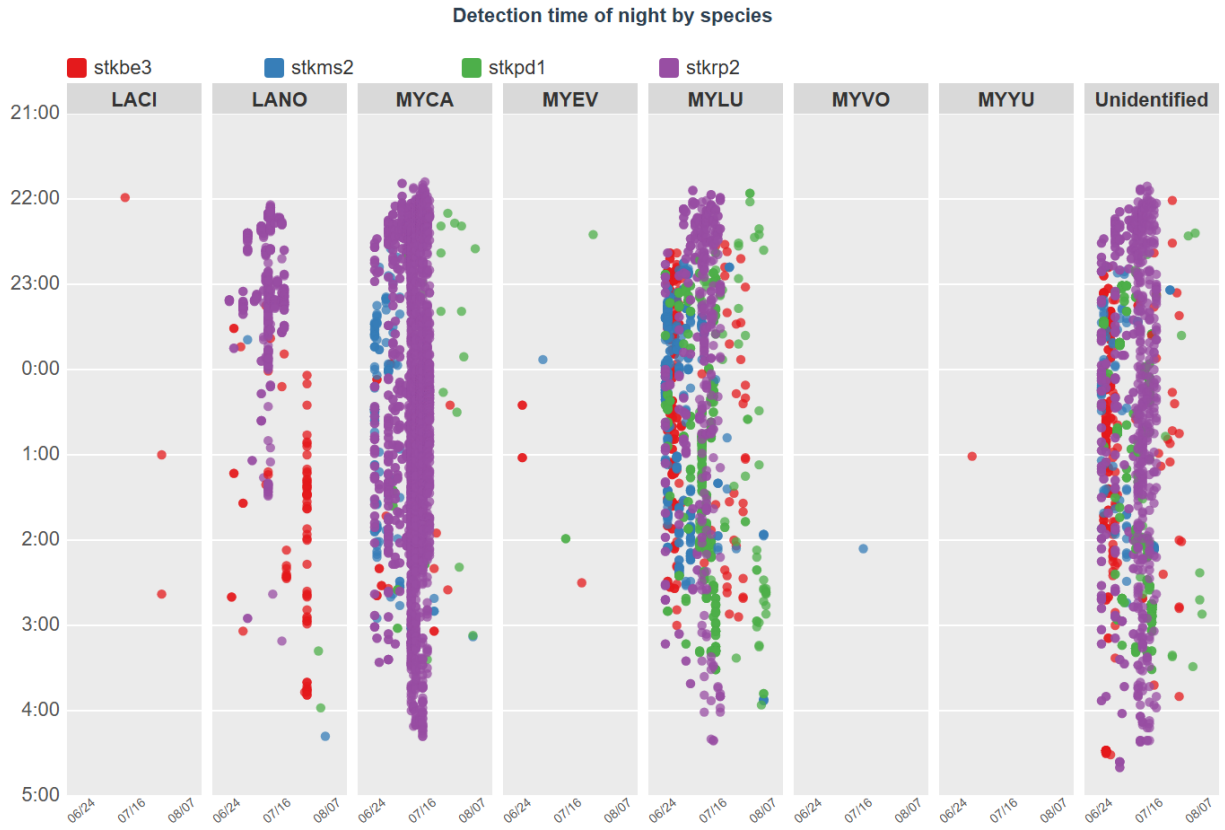
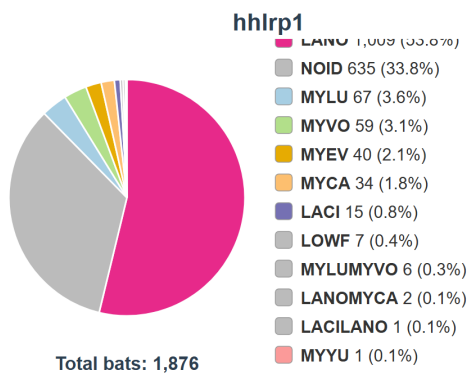
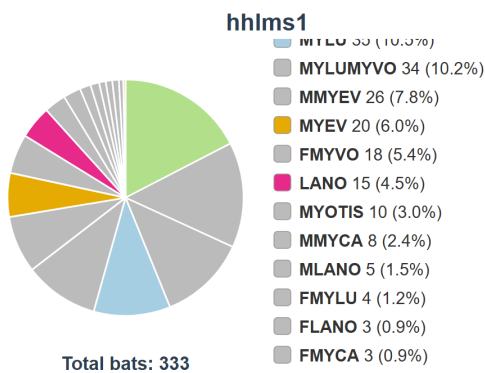
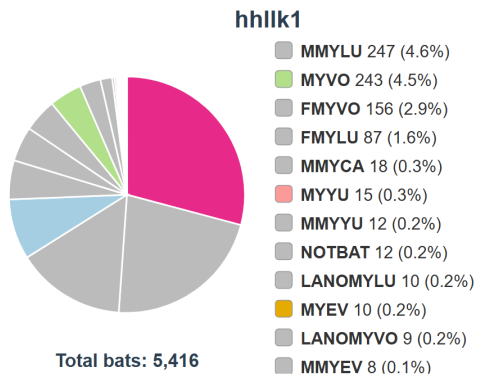
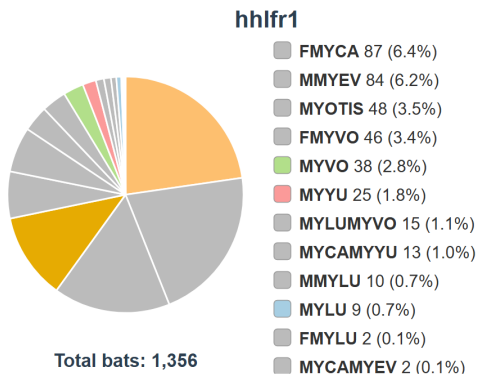


Figure 10: 2025 Detection times by species and site, cell 33486

Grid cell 47310



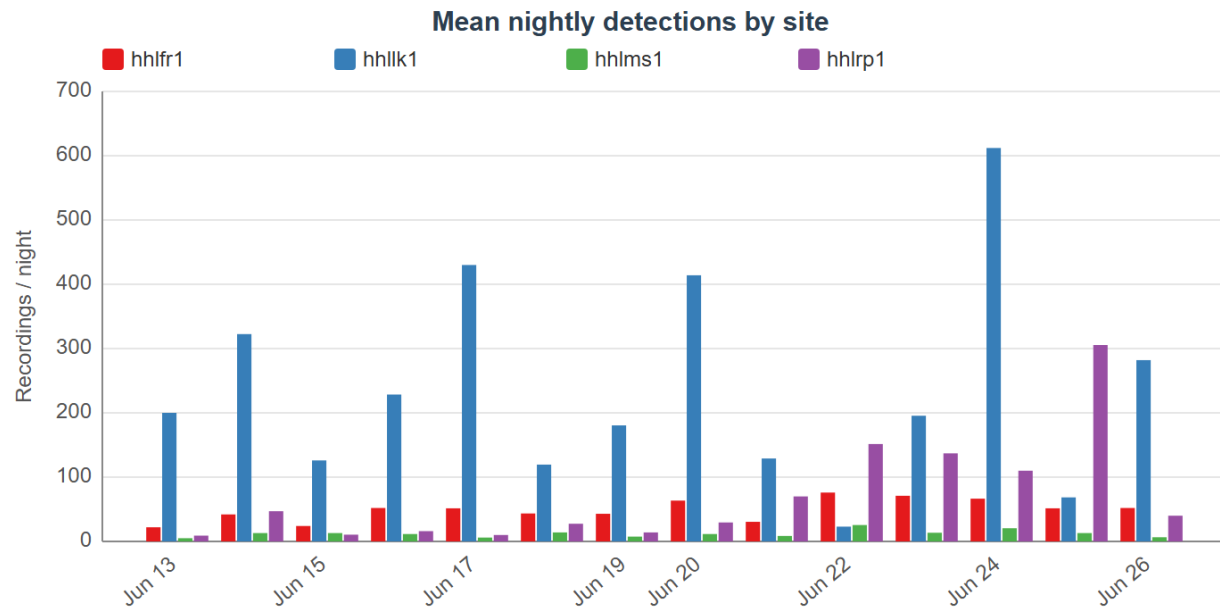


Figure 11: 2025 Nightly activity by site, cell 47310

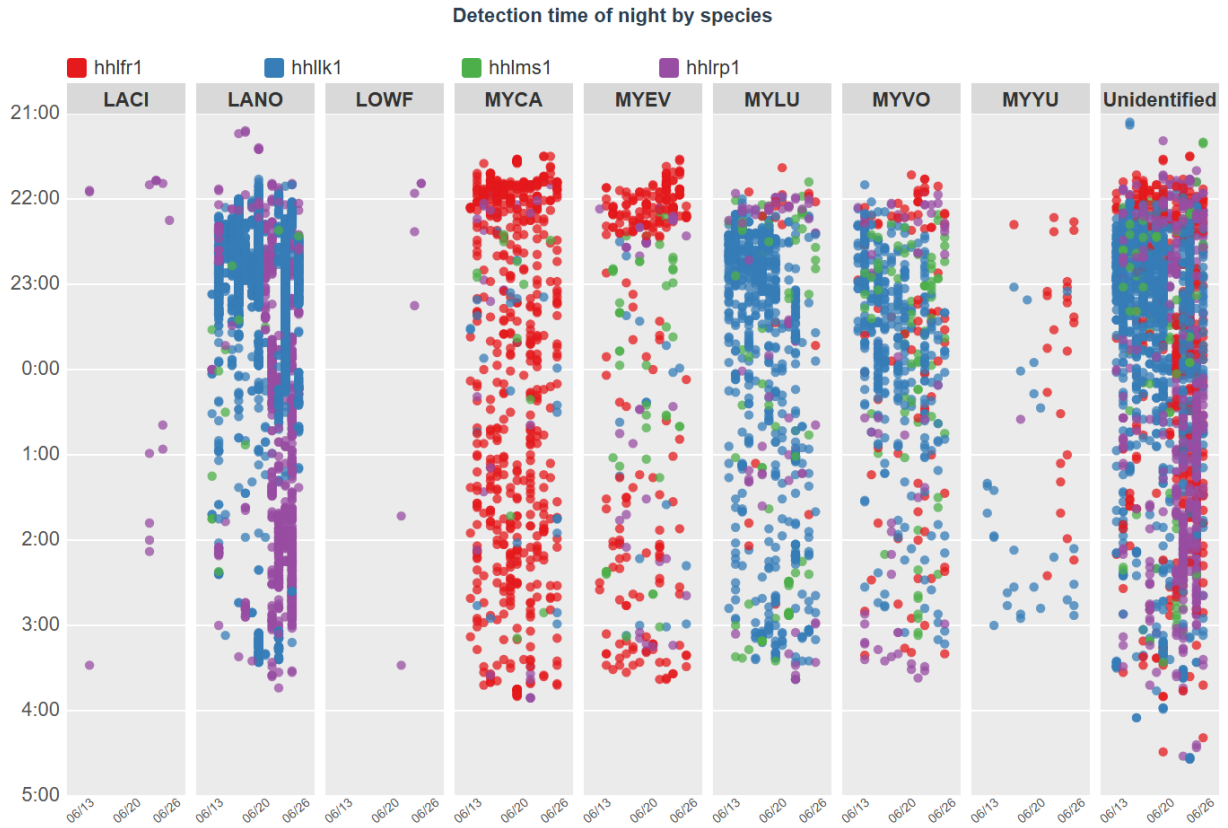


Figure 12: 2025 Detection times by species and site, cell 47310

Grid cell 117173

Jewel Lake

No detections

No sampling data

Figure 13: 2025 Nightly activity by site, cell 117173

No detections

Figure 14: 2025 Detection times by species and site, cell 117173

Grid cell 154037

Little Campbell Lake

No detections

No sampling data

Figure 15: 2025 Nightly activity by site, cell 154037

No detections

Figure 16: 2025 Detection times by species and site, cell 154037

Grid cell 203189

Basher Lake

No detections

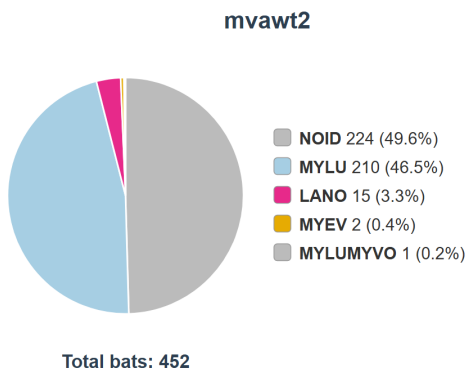
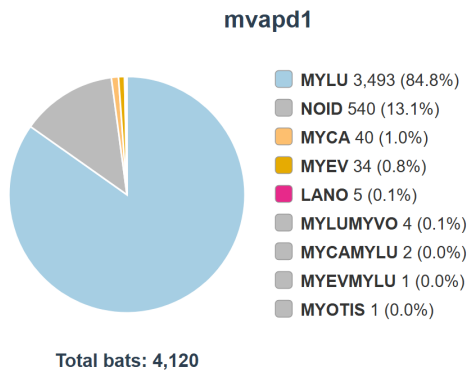
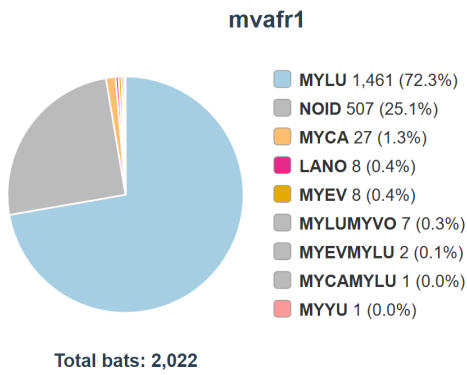
No sampling data

Figure 17: 2025 Nightly activity by site, cell 203189

No detections

Figure 18: 2025 Detection times by species and site, cell 203189

Grid cell 315563



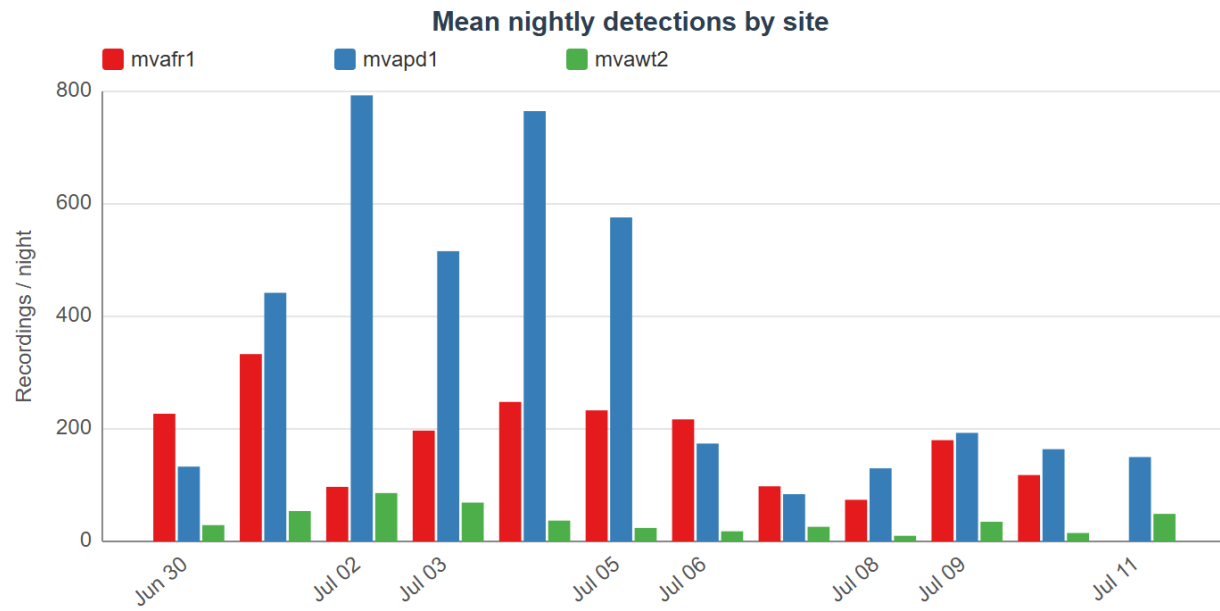


Figure 19: 2025 Nightly activity by site, cell 315563

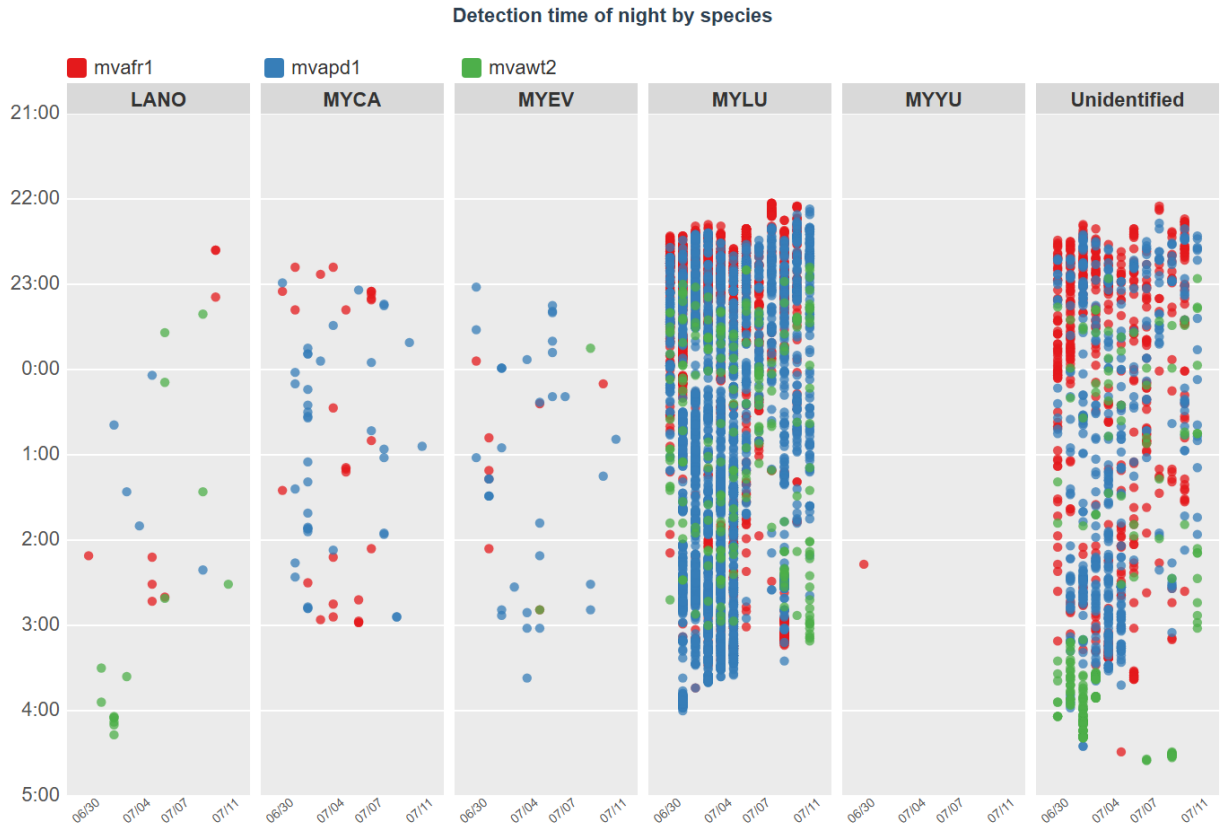


Figure 20: 2025 Detection times by species and site, cell 315563

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